

Algebra 2 Final Exam Study Guide**Multiple Choice**

Identify the choice that best completes the statement or answers the question.

_____ 1. Solve $|6y - 4| = 18$.

a. $x = -2\frac{1}{3}, x = 3\frac{2}{3}$

c. $x = 2\frac{1}{3}, x = -3\frac{2}{3}$

b. $x = -3\frac{2}{3}, x = 3\frac{2}{3}$

d. $x = 2\frac{1}{3}, x = 3\frac{2}{3}$

_____ 2. Solve $2|2x + 4| - 1 = 7$.

a. $x = 0$

c. $x = 0, x = 1$

b. $x = -4, x = 0$

d. $x = -4, x = 1$

_____ 3. Solve $|3n - 24| \leq 9$.

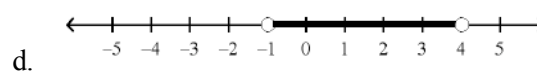
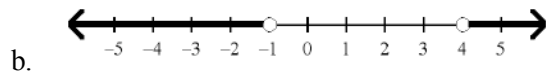
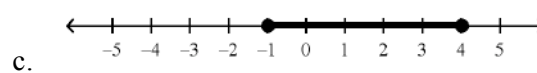
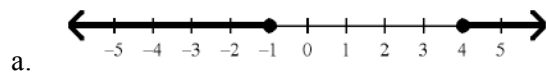
a. $n \leq 5$ or $n \leq 11$

c. $n \leq 5$ or $n \geq 11$

b. $n \leq 11$

d. $5 \leq n \leq 11$

_____ 4. Graph the solution for $|6x - 9| > 15$.



_____ 5. Solve $2|x - 3| = -1$.

a. $x = -3\frac{1}{2}, 2\frac{1}{2}$

c. $x = -3\frac{1}{2}, -2\frac{1}{2}$

b. $x = 2\frac{1}{2}, 3\frac{1}{2}$

d. no solution

_____ 6. Solve the system of equations.

$$2x + 5y = 26$$

$$6x - y = 14$$

a. $(-2, 6)$

c. infinite number of solutions

b. $(3, 4)$

d. no solution

_____ 7. Solve the system of equations.

$$3x - 2y = 24$$

$$-9x + 6y = -18$$

a. $(10, 3)$

c. infinite number of solutions

b. $(2, -9)$

d. no solution

_____ 8. Solve the system of equations.

$$x + 3y - 2z = -9$$

$$6x - 4y + 3z = 29$$

$$2x + 2y + z = 13$$

a. $(2, 1, 7)$

b. $(3, 2, 9)$

c. infinite number of solutions

d. no solution

_____ 9. Which polynomial represents $(x^3 + 4x^2 - 7) - (5x^3 - 2x^2 + 2x - 16)$?

a. $6x^3 + 2x^2 - 2x + 9$

b. $-4x^3 + 6x^2 + 9$

c. $-4x^3 + 2x^2 - 2x + 9$

d. $-4x^3 + 6x^2 - 2x + 9$

_____ 10. Which polynomial represents $(2x - 5)(6x^2 + 8)$?

a. $-18x^2 + 16x - 40$

b. $42x^2 + 16x - 40$

c. $12x^3 - 40$

d. $12x^3 - 30x^2 + 16x - 40$

_____ 11. $(9x^2 - 4x + 8)(3x^2 - 6x + 2)$

a. $27x^4 - 66x^3 + 66x^2 - 56x + 16$

b. $27x^2 + 24x + 16$

c. $27x^4 - 66x^3 + 42x^2 - 64x + 16$

d. $6x^2 - 10x + 10$

_____ 12. $(x^2 - 6x - 16) \div (x + 2)$

a. $x - 3$

b. $x - 4$

c. $x - 8$

d. $x + 8$

_____ 13. $3x - 2 \overline{) 9x^4 - 3x^3 + 16x^2 - 36x + 16}$

a. $3x^4 + x^3 + 6x^2 - 8x$

b. $3x^3 + x^2 + 6x - 8$

c. $3x^3 - x^2 + 6x - 8$

d. $3x^3 + x^2 + 6x + \frac{32}{3x - 2}$

_____ 14. Which polynomial represents $(x - 5)(x + 4)(x^2 - 1)$?

a. $x^4 - x^3 - 22x^2 - x + 20$

b. $x^4 + 20$

c. $x^4 - 22x^2 + x + 20$

d. $x^4 - x^3 - 21x^2 + x + 20$

_____ 15. $2x + 4 \overline{) 12x^4 + 30x^3 - 6x^2 + 36x + 8}$

a. $6x^3 + 3x^2 - 9x + 36$

b. $6x^4 + 3x^3 - 9x^2 + 36x - 136$

c. $6x^3 + 3x^2 - 9x - 100$

d. $6x^3 + 3x^2 - 9x + 36 - \frac{136}{2x + 4}$

_____ 16. $16x^2 - 121 =$

a. $(4x + 11)(4x + 11)$

b. $(4x + 11)(4x - 11)$

c. $(4x + 11)^2$

d. $(4x - 11)^2$

_____ 17. $4x^2 + 20x + 25 =$

a. $(4x + 25)^2$

b. $(2x + 5)(2x - 5)$

c. $(2x + 5)^2$

d. $(2x + 10)^2$

_____ 18. $x^3 + 27 =$

a. $(x + 3)(x^2 - 3x + 9)$

b. $(x - 3)(x^2 + 3x + 9)$

c. $(x + 3)^3$

d. $(x + 3)^2(x - 3)$

_____ 19. $27x^3 - 64 =$

a. $(3x - 4)(9x^2 + 12x + 16)$

b. $(3x + 4)(9x^2 - 12x + 16)$

c. $(3x - 4)(3x + 4)^2$

d. $(3x - 4)^3$

_____ 20. A square has an area of $16x^2 + 24x + 9$. What is the length of a side of the square?

a. $(4x + 3)^2$

b. $(4x - 3)$

c. $(8x + 3)$

d. $(4x + 3)$

_____ 21. If $i = \sqrt{-1}$, then $\sqrt{-25} + 3 =$

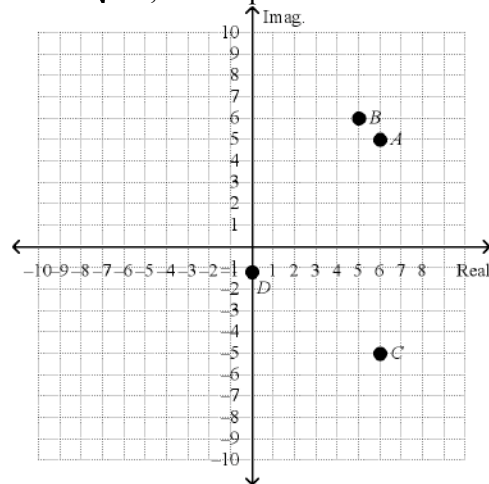
a. $3 + 5\sqrt{i}$

b. -5

c. $3 + 5i$

d. $3 - 5i$

_____ 22. If $i = \sqrt{-1}$, which point shows the location of $6 + 5i$ on the plane?



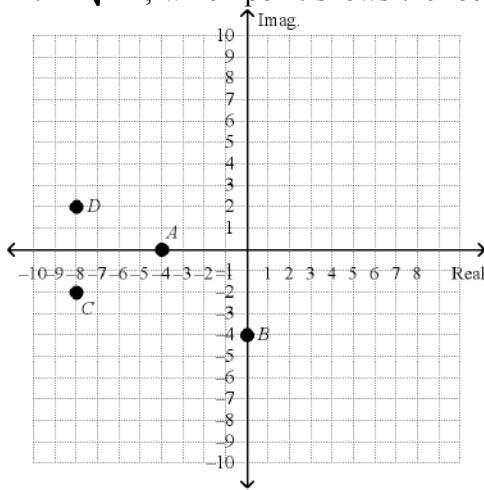
a. point A

b. point B

c. point C

d. point D

_____ 23. If $i = \sqrt{-1}$, which point shows the location of $-8 - 2i$ on the plane?



- a. point A
b. point B

- c. point C
d. point D

_____ 24. $(-4 + 3i) - (-6 - i) =$

- a. $2 + 2i$
b. $2 + 4i$

- c. $-10 + 2i$
d. $-10 + 4i$

_____ 25. $(-7 + 8i) + (-1 - 3i) =$

- a. $-8 + 5i$
b. $-6 + 5i$

- c. $-8 + 11i$
d. $-6 + 11i$

_____ 26. If $i = \sqrt{-1}$, then $(12i)(9i) =$

- a. $108i$
b. $108\sqrt{-1}$

- c. -108
d. 108

_____ 27. What is an equivalent form of $\frac{6}{4 + 2i}$?

- a. 3

c. $\frac{2 + i}{3}$

- b. $2 - i$

d. $\frac{6 - 3i}{5}$

_____ 28. If $i = \sqrt{-1}$, then $\frac{6 + \sqrt{-16}}{2 + \sqrt{-4}} =$

- a. $3 + 2i$

c. $\frac{1 + 4i}{3}$

- b. $3 - 4i$

d. $\frac{5 - i}{2}$

_____ 29. Which is a simplified form of $\frac{x^2 - x - 12}{x^2 - 16}$?

a. $\frac{x + 3}{x + 4}$

c. $\frac{x + 3}{4}$

b. $\frac{x + 3}{x - 4}$

d. $\frac{3}{4}$

_____ 30. $\frac{2x - 10}{x^2 - 3x - 10} \cdot \frac{x^2 - 4}{8x - 16} =$

a. $\frac{16}{(x + 2)^2}$

c. $\frac{1}{8}$

b. $\frac{x - 2}{4(x + 2)}$

d. $\frac{1}{4}$

_____ 31. $\frac{4x + 20}{x^2 - 1} \div \frac{2x + 10}{x^2 + 4x - 5} =$

a. $\frac{4(x + 5)}{(x + 1)}$

c. $\frac{2(x + 5)}{(x + 1)(x - 1)^2}$

b. $\frac{2(x + 5)}{(x + 1)}$

d. $\frac{2(x + 1)}{4x}$

_____ 32. $\frac{8}{3x + 6} + \frac{6x}{x^2 + x - 2} =$

a. $\frac{26x - 8}{3x^2 + 3x - 6}$

c. $\frac{48x}{3x^3 + 9x^2 - 12}$

b. $\frac{8 + 6x}{x^2 + 4x + 4}$

d. $\frac{14x - 8}{3x^2 + 3x - 6}$

_____ 33. $\frac{x}{3x - 3} - \frac{x + 1}{x^2 - 1} =$

a. $\frac{1}{x^2 - 3x - 2}$

c. $\frac{1}{3}$

b. $\frac{x - 3}{3x - 3}$

d. $\frac{x - 1}{3x - 3}$

_____ 34. Which is a simplified form of $\frac{5x^{-2}y^6z}{15x^3y^4z^{-2}}$?

a. $\frac{y^2z^3}{3x^5}$

c. $\frac{y^2}{3xz^{-1}}$

b. $\frac{xy^2z}{3}$

d. $\frac{x^5y^{10}z^3}{3}$

_____ 35. Which is a simplified form of $\frac{(2a^2b^{-2})^3}{4b^3c^{-8}}$?

a. $\frac{3a^5c^8}{2b^2}$

c. $\frac{a^6c^8}{2b^9}$

b. $\frac{2a^6}{b^3c^8}$

d. $\frac{2a^6c^8}{b^9}$

_____ 36. $\frac{6x+18}{4x-8} \cdot \frac{3x-2}{x+3} =$

a. $\frac{3}{2}$

c. $\frac{9x-6}{2x-4}$

b. $\frac{9}{2}$

d. $\frac{9x^2-18}{2x^2-12}$

_____ 37. $\frac{1}{x+3} + \frac{x}{x+4} =$

a. $\frac{1+x}{2x+7}$

c. $\frac{1}{x+3}$

b. $\frac{x^2+4x+4}{x^2+7x+12}$

d. $\frac{x+2}{x+4}$

_____ 38. What are the solutions to the equation $x^2 - 8x - 48$?

a. $x = -8; x = 6$

c. $x = -4; x = 12$

b. $x = -12; x = 4$

d. $x = -16; x = 3$

_____ 39. What are the solutions to the equation $2x^2 + 7x - 15$?

a. $x = -\frac{3}{2}; x = 5$

c. $x = -3; x = \frac{5}{2}$

b. $x = -5; x = \frac{3}{2}$

d. $x = -\frac{5}{2}; x = 3$

_____ 43. What are the solutions to the equation $x^2 + 6x + 27$?

- a. $x = -3 + 3i\sqrt{2}; x = -3 - 3i\sqrt{2}$
- b. $x = -3 + 3\sqrt{2}; x = -3 - 3\sqrt{2}$
- c. $x = -9; x = 3$
- d. $x = -3; x = 9$

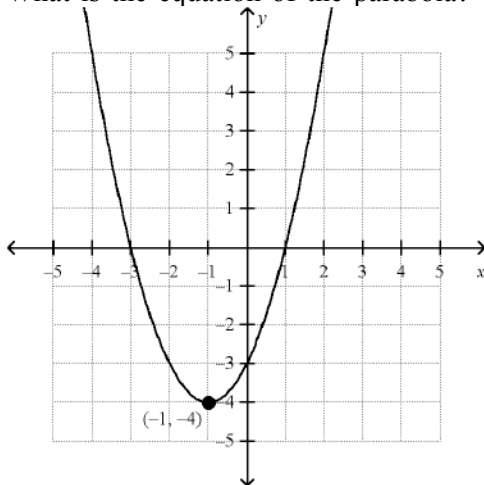
_____ 44. What are the solutions to the equation $\frac{4}{x^2} + \frac{9}{x} = -2$?

- a. $x = -4; x = -\frac{1}{2}$
- b. $x = -2; x = -\frac{1}{4}$
- c. $x = -\frac{9}{4} + \frac{\sqrt{113}}{4}; x = -\frac{9}{4} - \frac{\sqrt{113}}{4}$
- d. $x = \frac{9}{4} + \frac{\sqrt{113}}{4}; x = \frac{9}{4} - \frac{\sqrt{113}}{4}$

_____ 45. What is the equation of a parabola that is translated down 12 and 15 to the left from the graph of the parent function $y = x^2$?

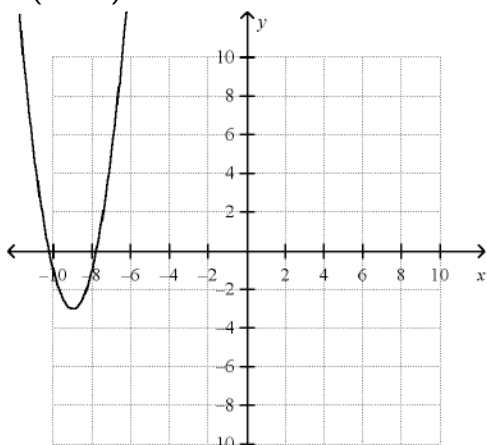
- a. $y = (x + 15)^2 - 12$
- b. $y = (x - 15)^2 - 12$
- c. $y = (x + 12)^2 - 15$
- d. $y = (x - 12)^2 - 15$

_____ 46. What is the equation of the parabola?

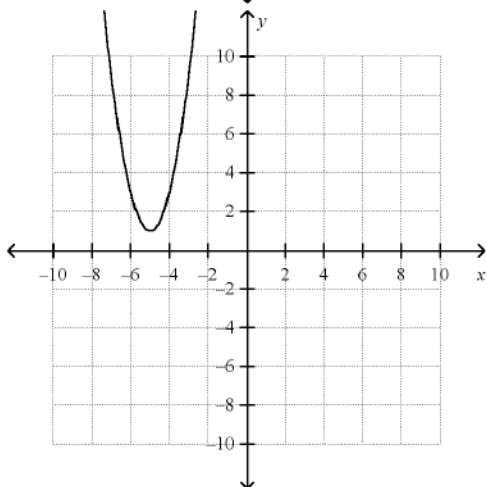


- a. $y = (x - 1)^2 - 4$
- b. $y = (x + 1)^2 - 4$
- c. $y = (x + 1)^2 + 4$
- d. $y = (x - 1)^2 + 4$

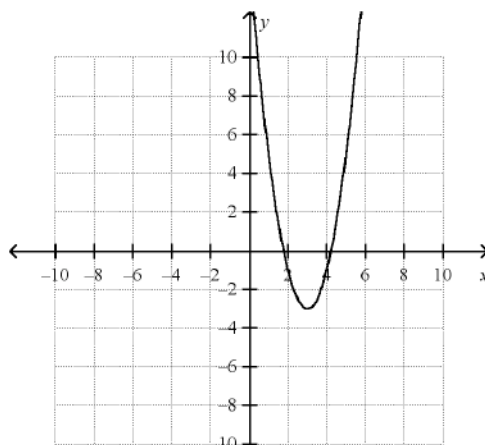
- _____ 47. Which of the following *most* accurately describes the translation of the graph $y = (x - 5)^2 + 7$ to the graph of $y = (x + 1)^2 - 4$?
- a. down 11 and 6 to the left c. down 3 and 4 to the left
b. down 11 and 6 to the right d. down 3 and 4 to the right
- _____ 48. What is the equation of a parabola that is translated down 6 and 4 to the left from the graph of the parabola $y = (x - 2)^2 + 4$?
- a. $y = (x + 4)^2$ c. $y = (x + 4)^2 + 8$
b. $y = (x + 2)^2 - 2$ d. $y = (x - 2)^2 - 2$
- _____ 49. Which of the following *most* accurately describes the translation of the graph $y = x^2 + 12x + 44$ to the graph of $y = x^2 - 6x + 7$?
- a. down 37 and 18 to the right c. down 9 and 10 to the right
b. up 37 and 18 to the left d. down 10 and 9 to the right
- _____ 50. What is the graph of the parabola that is translated up 6 and 2 to the right of the graph of $y = 2(x + 3)^2 - 5$?



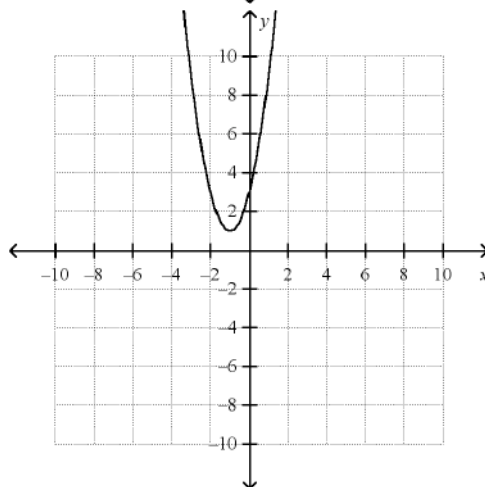
a.



b.

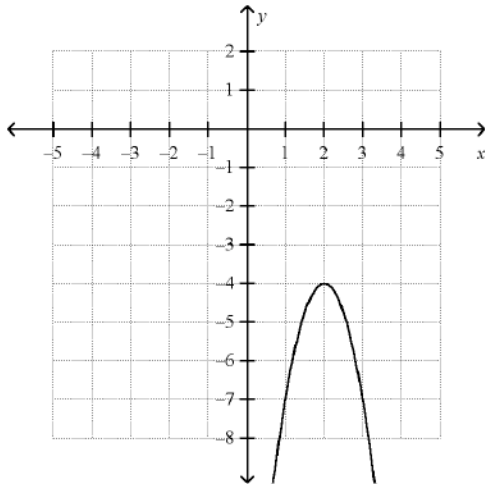


c.

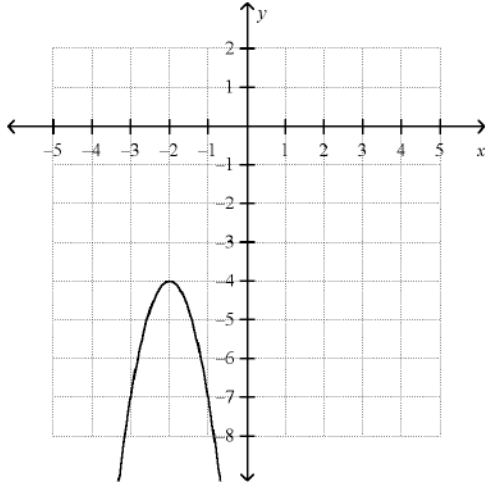


d.

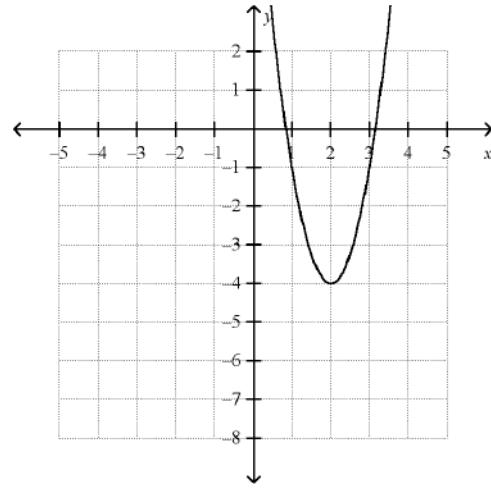
_____ 51. Which is the graph of $y = -3(x + 2)^2 - 4$?



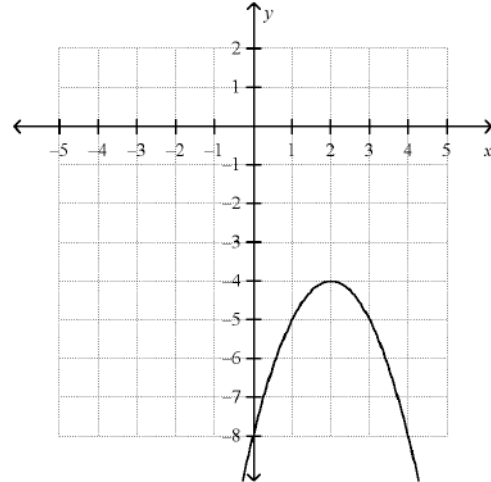
a.



b.



c.



d.

_____ 52. Which of these equations is true?

a. $\log_5 2 = 25$

c. $\log_5 25 = 2$

b. $\log_2 5 = 25$

d. $\log_2 25 = 5$

_____ 53. $\frac{\log_4 7}{\log_4 16} =$

a. $\frac{16}{7} \log 4$

c. $\log_{16} 7$

b. $\log_7 16$

d. $\log_4 \frac{7}{16}$

_____ 54. What is the value of $\log_2 8 + \log_2 4$?

a. $\log_2 12$

c. 6

b. 5

d. 12

- _____ 55. What is the solution to the equation $4^{3x} = 97$?
- a. $x = 8.08$ c. $x = \log 97$
b. $x = \frac{\log_{10} 97 - \log_{10} 4}{3}$ d. $x = \frac{\log_{10} 97}{3 \log_{10} 4}$
- _____ 56. $\log_2 10 =$
- a. $\frac{\log_8 10}{\log_8 2}$ c. $10 \log 2$
b. $\frac{\log_7 2}{\log_7 10}$ d. $\log_5 10 - \log_5 2$
- _____ 57. If $\log_4 (7x - 34) = 6$, what is the value of x ?
- a. $585\frac{1}{7}$ c. 590
b. $587\frac{6}{7}$ d. 610
- _____ 58. Which of these is *not* equivalent to $\log_3 9^5$?
- a. $5 \log_3 9$ c. $\log_3 9 + \log_3 9 + \log_3 9 + \log_3 9 + \log_3 9$
b. 10 d. 2^5
- _____ 59. Which of these equations is true?
- a. $\log_3 2 = 8$ c. $\log_2 8 = 3$
b. $\log_2 3 = 8$ d. $\log_3 8 = 2$
- _____ 60. What is the solution to the equation $2^x = 5$?
- a. $x = 2.5$ c. $x = \log 5$
b. $x = \frac{\log_{10} 5}{\log_{10} 2}$ d. $x = \frac{\log_{10} 2}{\log_{10} 5}$
- _____ 61. What is the solution to the equation $4^x = 7$?
- a. $x = \frac{\log_{10} 7}{\log_{10} 4}$ c. $x = \log \frac{7}{4}$
b. $x = \frac{\log_{10} 4}{\log_{10} 7}$ d. $x = \frac{4}{3}$

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_____ 62. If $\log (5x + 20) = 3$, what is the value of x ?

a. $-\frac{19}{5}$

c. 196

b. $-\frac{17}{5}$

d. 204

_____ 63. If $\log (12x + 40) = 2$, what is the value of x ?

a. $-\frac{19}{6}$

c. 80

b. 5

d. 82

_____ 64. $\frac{3^{\frac{2}{5}}}{3^{\frac{4}{5}}} =$

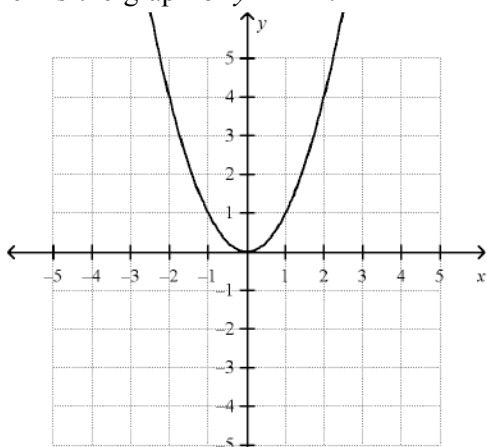
a. $3^{\frac{1}{2}}$

c. $\frac{1}{3^{\frac{2}{5}}}$

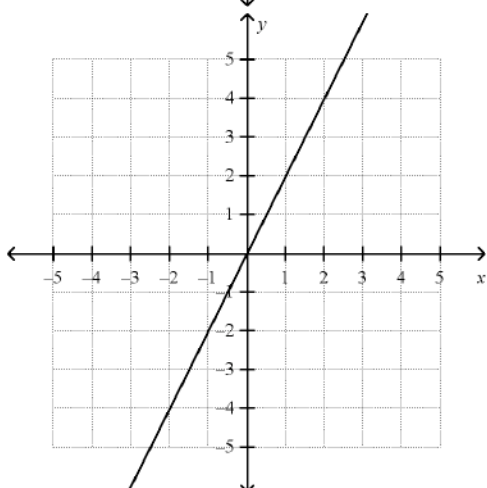
b. $\frac{1}{3^{-\frac{2}{5}}}$

d. $3^{\frac{6}{5}}$

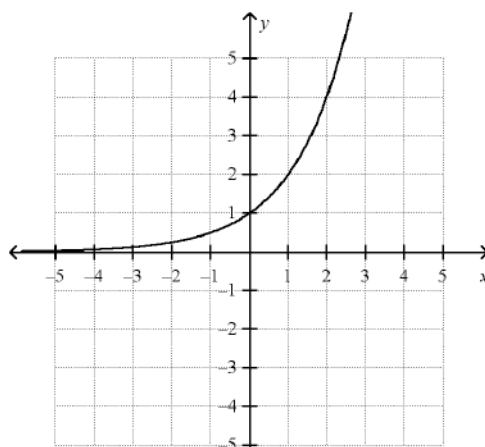
_____ 65. Which is the graph of $y = 2^x$?



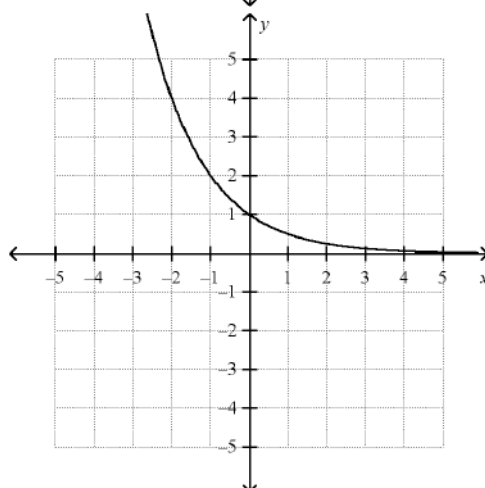
a.



b.



c.



d.

_____ 66. $\log_2 18 =$

a. $\log_{10} 9$

b. $18 \log 2$

c. $\frac{\log_{10} 2}{\log_{10} 18}$

d. $\frac{\log_{10} 18}{\log_{10} 2}$

_____ 67. $\log_5 4 =$

a. $\frac{\log_6 4}{\log_6 5}$

b. $\frac{\log_3 5}{\log_3 4}$

c. $4 \log 5$

d. $\log_{10} 4 - \log_{10} 5$

_____ 68. $\log_x 2 =$

- | | |
|----------------------|----------------------------|
| a. $\log 2 - \log x$ | c. $\frac{\log x}{\log 2}$ |
| b. $2 \log x$ | d. $\frac{\log 2}{\log x}$ |

_____ 69. $8! =$

- | | |
|-------|---------------|
| a. 8 | c. 40,320 |
| b. 64 | d. 16,777,216 |

_____ 70. ${}_6P_5 =$

- | | |
|--------|--------|
| a. 6 | c. 360 |
| b. 144 | d. 720 |

_____ 71. ${}_{10}C_5 =$

- | | |
|---------|------------|
| a. 252 | c. 30,240 |
| b. 1260 | d. 145,152 |

_____ 72. A pizza restaurant offers 8 toppings. How many 4-topping pizzas are possible?

- | | |
|-------|---------|
| a. 24 | c. 1680 |
| b. 70 | d. 2520 |

_____ 73. A bag has four marbles: one red, one yellow, one blue, and one green. Two marbles are taken from the bag. What is the probability that the two marbles are red and green?

- | | |
|-------------------|------------------|
| a. $\frac{1}{12}$ | c. $\frac{1}{4}$ |
| b. $\frac{1}{6}$ | d. $\frac{1}{2}$ |

_____ 74. Jackson has homework in 5 of his classes: Spanish, History, Math, English, and Science. If he randomly chooses the order in which to study, what is the probability that he does Math first, Science second, and Spanish third?

- | | |
|-------------------|-------------------|
| a. $\frac{1}{10}$ | c. $\frac{1}{30}$ |
| b. $\frac{1}{20}$ | d. $\frac{1}{60}$ |

_____ 75. Ten cars numbered 1 to 10 are racing. What is the probability that car number 5 wins the race and car number 8 comes in second?

- | | |
|-------------------|--------------------------|
| a. $\frac{1}{5}$ | c. $\frac{1}{90}$ |
| b. $\frac{1}{45}$ | d. $\frac{1}{1,814,400}$ |

- _____ 76. $(x + 8)^4 =$
- a. $x^4 + 4096$
 - b. $x^4 + 32x^3 + 384x^2 + 2048x + 4096$
 - c. $x^4 + 32x^3 + 384x^2 + 32x + 8$
 - d. $x^4 + 32x^3 + 384x^2 + 2048x + 8$
- _____ 77. $(2x + 3)^3 =$
- a. $8x^3 - 27$
 - b. $8x^3 + 36x^2 + 54x + 27$
 - c. $2x^3 + 18x^2 + 54x + 27$
 - d. $2x^3 + 36x^2 + 54x + 27$
- _____ 78. What is the fifth term of the expansion of $(3x + 2)^8$?
- a. $90,720x^4$
 - b. $72,576x^4$
 - c. $3,360x^4$
 - d. $420x^4$
- _____ 79. What is the fourth term of the expansion of $(2x - 1)^5$?
- a. $10x$
 - b. $-10x$
 - c. 1
 - d. -1
- _____ 80. What is the third term of the expansion of $(x + 2)^{12}$?
- a. $66x^{10}$
 - b. $528x^9$
 - c. $264x^9$
 - d. $264x^{10}$
- _____ 81. $f(x) = -x^2 + 2x$ and $g(x) = x^2 - 4$. What is $(f - g)(x)$?
- a. $-2x^2 + 2x + 4$
 - b. $2x^2 - 2x - 4$
 - c. $-2x^2 + 6x$
 - d. $2x + 4$
- _____ 82. $f(x) = x^2 + 1$ and $g(x) = 2x - 2$. What is $(f \cdot g)(x)$?
- a. x^2
 - b. $x^2 + 2x - 2$
 - c. $4x^2 - 8x + 5$
 - d. $2x^3 - 2x^2 + 2x - 2$

_____ 83. What is the inverse of $y = 2x - 3$?

a. $y = -2x - 3$

c. $y = \frac{1}{2}x - \frac{3}{2}$

b. $x = 2y - 3$

d. $y = \frac{1}{2}x + \frac{3}{2}$

_____ 84. What is the inverse of $f(x) = \sqrt{x - 8}$?

a. $f^{-1}(x) = -\sqrt{x - 8}$

c. $f^{-1}(x) = x^2 + 8, x \geq 0$

b. $f^{-1}(x) = x^2 + 8$

d. $f^{-1}(x) = x^2 + 8, x \geq 8$

_____ 85. If $f(x) = 2x^2 - 6x + 4$ and $g(x) = 4(x - 2)^2$, which is an equivalent form of $f(x) + g(x)$?

a. $6x^2 - 22x + 20$

c. $6x^2 - 6x - 12$

b. $18x^2 - 70x + 68$

d. $3x^2 - 10x + 8$

_____ 86. If $f(x) = x^5$ and $g(x) = 2x^3$, which is an equivalent form of $f(x) \cdot g(x)$?

a. $32x^{243}$

c. $2x^8$

b. $32x^{15}$

d. $2x^{15}$

_____ 87. If $f(x) = x^2 + 2x - 4$ and $g(x) = 2x - 4$, which is an equivalent form of $\frac{f(x)}{g(x)}$?

a. x^2

c. $\frac{x^2}{2x - 4}$

b. $\frac{x + 2}{2}$

d. $\frac{x^2 + 2x - 4}{2x - 4}$

_____ 88. If $\log 4 \approx 0.602$ and $\log 5 \approx 0.699$, what is the approximate value of $\log 400$?

a. 0.421

c. 1.301

b. 1.161

d. 2.602

_____ 89. If $\log_4 3 \approx 0.792$, $\log_4 4 = 1$, and $\log_4 5 \approx 1.161$, what is the approximate value of $\log_4 60$?

a. 2.953

c. 1.176

b. 1.778

d. 0.920

_____ 90. Which of these is *not* equivalent to $\log_4 \frac{45}{9}$?

a. $\frac{\log_4 45}{\log_4 9}$

c. $\log_4 45 - \log_4 9$

b. $\log_4 5$

d. $\frac{\log 5}{\log 4}$

Algebra 2 Final Exam Study Guide

Answer Section

MULTIPLE CHOICE

1. ANS: A	PTS: 1	STA: CA A2 1.0
2. ANS: B	PTS: 1	STA: CA A2 1.0
3. ANS: D	PTS: 1	STA: CA A2 1.0
4. ANS: B	PTS: 1	STA: CA A2 1.0
5. ANS: D	PTS: 1	STA: CA A2 1.0
6. ANS: B	PTS: 1	STA: CA A2 2.0
7. ANS: D	PTS: 1	STA: CA A2 2.0
8. ANS: A	PTS: 1	STA: CA A2 2.0
9. ANS: D	PTS: 1	STA: CA A2 3.0
10. ANS: D	PTS: 1	STA: CA A2 3.0
11. ANS: A	PTS: 1	STA: CA A2 3.0
12. ANS: C	PTS: 1	STA: CA A2 3.0
13. ANS: B	PTS: 1	STA: CA A2 3.0
14. ANS: D	PTS: 1	STA: CA A2 3.0
15. ANS: D	PTS: 1	STA: CA A2 3.0
16. ANS: B	PTS: 1	STA: CA A2 4.0
17. ANS: C	PTS: 1	STA: CA A2 4.0
18. ANS: A	PTS: 1	STA: CA A2 4.0
19. ANS: A	PTS: 1	STA: CA A2 4.0
20. ANS: D	PTS: 1	STA: CA A2 4.0
21. ANS: C	PTS: 1	STA: CA A2 5.0
22. ANS: A	PTS: 1	STA: CA A2 5.0
23. ANS: C	PTS: 1	STA: CA A2 5.0
24. ANS: B	PTS: 1	STA: CA A2 6.0
25. ANS: A	PTS: 1	STA: CA A2 6.0
26. ANS: C	PTS: 1	STA: CA A2 6.0
27. ANS: D	PTS: 1	STA: CA A2 6.0
28. ANS: D	PTS: 1	STA: CA A2 6.0
29. ANS: A	PTS: 1	STA: CA A2 7.0
30. ANS: D	PTS: 1	STA: CA A2 7.0
31. ANS: B	PTS: 1	STA: CA A2 7.0
32. ANS: A	PTS: 1	STA: CA A2 7.0
33. ANS: B	PTS: 1	STA: CA A2 7.0
34. ANS: A	PTS: 1	STA: CA A2 7.0
35. ANS: D	PTS: 1	STA: CA A2 7.0
36. ANS: C	PTS: 1	STA: CA A2 7.0
37. ANS: B	PTS: 1	STA: CA A2 7.0
38. ANS: C	PTS: 1	STA: CA A2 8.0
39. ANS: B	PTS: 1	STA: CA A2 8.0
40. ANS: D	PTS: 1	STA: CA A2 8.0

41.	ANS: B	PTS: 1	STA: CA A2 8.0
42.	ANS: B	PTS: 1	STA: CA A2 8.0
43.	ANS: A	PTS: 1	STA: CA A2 8.0
44.	ANS: A	PTS: 1	STA: CA A2 8.0
45.	ANS: A	PTS: 1	STA: CA A2 9.0
46.	ANS: B	PTS: 1	STA: CA A2 9.0
47.	ANS: A	PTS: 1	STA: CA A2 9.0
48.	ANS: B	PTS: 1	STA: CA A2 9.0
49.	ANS: D	PTS: 1	STA: CA A2 9.0
50.	ANS: D	PTS: 1	STA: CA A2 9.0
51.	ANS: B	PTS: 1	STA: CA A2 10.0
52.	ANS: C	PTS: 1	STA: CA A2 11.0
53.	ANS: C	PTS: 1	STA: CA A2 11.0
54.	ANS: B	PTS: 1	STA: CA A2 11.0
55.	ANS: D	PTS: 1	STA: CA A2 11.0
56.	ANS: A	PTS: 1	STA: CA A2 11.0
57.	ANS: C	PTS: 1	STA: CA A2 11.0
58.	ANS: D	PTS: 1	STA: CA A2 11.0
59.	ANS: C	PTS: 1	STA: CA A2 11.1
60.	ANS: B	PTS: 1	STA: CA A2 11.1
61.	ANS: A	PTS: 1	STA: CA A2 11.1
62.	ANS: C	PTS: 1	STA: CA A2 11.1
63.	ANS: B	PTS: 1	STA: CA A2 11.1
64.	ANS: C	PTS: 1	STA: CA A2 12.0
65.	ANS: C	PTS: 1	STA: CA A2 12.0
66.	ANS: D	PTS: 1	STA: CA A2 13.0
67.	ANS: A	PTS: 1	STA: CA A2 13.0
68.	ANS: D	PTS: 1	STA: CA A2 13.0
69.	ANS: C	PTS: 1	STA: CA A2 18.0
70.	ANS: D	PTS: 1	STA: CA A2 18.0
71.	ANS: A	PTS: 1	STA: CA A2 18.0
72.	ANS: B	PTS: 1	STA: CA A2 18.0
73.	ANS: B	PTS: 1	STA: CA A2 19.0
74.	ANS: D	PTS: 1	STA: CA A2 19.0
75.	ANS: C	PTS: 1	STA: CA A2 19.0
76.	ANS: B	PTS: 1	STA: CA A2 20.0
77.	ANS: B	PTS: 1	STA: CA A2 20.0
78.	ANS: A	PTS: 1	STA: CA A2 20.0
79.	ANS: A	PTS: 1	STA: CA A2 20.0
80.	ANS: D	PTS: 1	STA: CA A2 20.0
81.	ANS: A	PTS: 1	STA: CA A2 24.0
82.	ANS: D	PTS: 1	STA: CA A2 24.0
83.	ANS: D	PTS: 1	STA: CA A2 24.0
84.	ANS: C	PTS: 1	STA: CA A2 24.0
85.	ANS: A	PTS: 1	STA: CA A2 25.0
86.	ANS: C	PTS: 1	STA: CA A2 25.0

87. ANS: D	PTS: 1	STA: CA A2 25.0
88. ANS: D	PTS: 1	STA: CA A2 14.0
89. ANS: A	PTS: 1	STA: CA A2 14.0
90. ANS: A	PTS: 1	STA: CA A2 14.0

Name: _____

Date: _____

Algebra II, Period ____

Math Department

Final Exam Review Packet - Algebra II

- This review packet contains questions that are similar to the type of problems that you will encounter on the exam.
 - The in-class review is not meant to re-teach you everything from the second semester. It will be a quick, but thorough overview of the material.
 - It is recommended that you work on this review packet leading up to your exam day so you have questions ready. Don't wait till the last minute.
 - Remember that the exam counts for 20% of your course grade.
 - Reviewing for the exam is **YOUR** responsibility.
 - If you have questions as you prepare, make arrangements to see your teacher.
-

I. Quadratics Equations

Solve each of the following equations using factoring.	
a. $x^2 - 36 = 0$	b. $7x^2 - 14x = 0$
c. $x^3 - 6x^2 - 7x = 0$	d. $6x^2 + 7x - 3 = 0$
e. $3x^2 + 3x - 36 = 0$	f. $32x^2 - 2 = 0$
g. $x^3 - 2x^2 - 9x + 18 = 0$	h. $x^3 - 3x^2 + 6x - 18 = 0$

Quadratics Equations (continued)

Factor each polynomial COMPLETELY.

Sum of Two Cubes:
$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

Difference of Two Cubes:
$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

a. $x^3 + 27$

b. $8x^3 - 125$

c. $x^4 + 5x^2 - 14$

d. $2x^5 - 18x^3 + 40x$

Solve each of the following equations using the Quadratic Formula.

Quadratic Formula:
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

a. $4x^2 + 6x + 1 = 0$

b. $x^2 + 2x + 2 = 0$

c. $2x^2 + 3x - 5 = 0$

d. $3x^2 - 2x - 7 = 0$

II. Powers, Roots, and Radicals

Rewrite the expression with positive exponents. Evaluate where possible.

a. $(-3)^{-4}$

b. $\frac{4}{x^0 + 7}$

c. $3x^3(2x)^2$

d. $\frac{8a^4b^6}{2(a^5b)^2}$

e. $4(x^{-3}y^4)(-3xy^2)^2$

f. $\frac{20(a^{-4}b^{-2})}{8(a^{-2}b^4)^{-2}}$

Solve the radical or rational exponent equation.

a. $x^{\frac{1}{5}} = 2$

b. $2\sqrt{3x-1} + 3 = 11$

c. $4x^2 = 64$

d. $2(x-2)^{\frac{1}{4}} - 3 = 159$

e. $\sqrt{2x+4} = \sqrt{x+2}$

f. $\sqrt[3]{x} - 6 = -2$

III. Simplifying Rational Expressions

Simplify the Rational Expression using Multiplication or Division.

a. $\frac{x^2 + 4x - 12}{x^2(x^2 + 9x + 18)} \cdot 6x^2$

f. $\frac{12x^2y^3z}{6x^3y^2z^2}$

b. $\frac{3x^2 - 12}{5x - 10} \cdot \frac{1}{2x + 4}$

g. $\frac{x^3 + 3x^2}{2x} \div \frac{x^2 + 5x + 6}{5x^3}$

c. $\frac{x^2 - 4}{x^2 + 4} \cdot \frac{x + 2}{x - 2}$

h. $\frac{x^2 + x - 20}{x + 1} \div \frac{11x + 55}{x + 1}$

d. $\frac{5x^2 - 20}{25x^2} \cdot \frac{x}{x - 2}$

i. $\frac{x^2 + 5x + 6}{x + 3} \div \frac{x^2 - 4}{x + 1}$

e. $x^2 + x - 30 \cdot \frac{x}{x^2 + 6x}$

j. $\frac{x^2 + 6x - 7}{3x^2} \div \frac{x + 7}{6x}$

Simplifying Rational Expressions (continued)

Simplify the Rational Expression using Addition or Subtraction. (LCD = ?)
--

a. $\frac{4}{3x^2} + \frac{2}{5x}$

b. $\frac{3}{2x-2} + \frac{x+1}{4}$

c. $\frac{4}{3x^3} + \frac{x}{6x^3 + 3x^2}$

d. $\frac{5x-1}{x^2+2x-8} - \frac{6}{x+4}$

e. $\frac{x+1}{x^2+6x+9} - \frac{1}{x^2-9}$

IV. Solving Rational Equations

Solve each rational equation.

a. $\frac{3}{x+4} = \frac{9}{x-2}$

b. $\frac{4x}{x-1} = \frac{x}{x^2-1}$

c. $\frac{3}{x^2-4} = \frac{2}{x+2} + \frac{x}{x-2}$

d. $\frac{3x-2}{x-2} = \frac{6}{x^2-4} + 1$

e. $\frac{x}{x+2} = \frac{3x+1}{x-1} + \frac{4}{x^2+x-2}$

V. Function Operations

Perform the indicated operation with the functions given.

Let $f(x) = x^2 - 3x + 4$, $g(x) = 5x + 2$, and $h(x) = 6x$.

a. $(f + g)(x) =$

b. $(f - h)(x) =$

c. $(g \cdot h)(x) =$

d. $(f + h)(-2) =$

e. $(h - g)(3) =$

f. $(g \cdot f)(0) =$

g. $(f \circ g)(x) =$

h. $(f \circ h)(x) =$

i. $(g \circ f)(1) =$

j. $(f \circ h)(-7) =$

k. $g(h(f(x))) =$

l. $f(g(h(-1))) =$

VI. Inverses

Find the inverse of each function.

a. $f(x) = 2x + 5$

b. $f(x) = \sqrt[3]{2x + 4}$

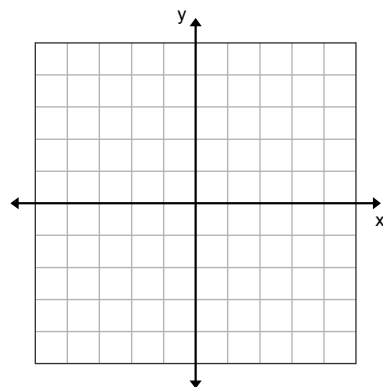
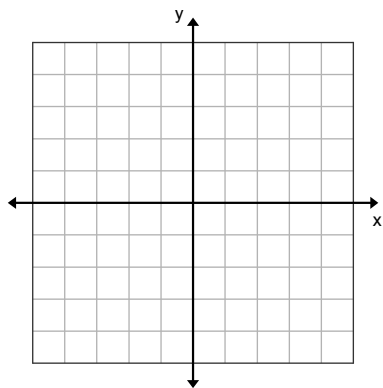
c. $f(x) = 5 - \frac{5}{2}x$

d. $f(x) = \frac{x-2}{4}$

Verify that the two functions are inverses of each other using composite functions. Then, verify (a) and (b) by graphing.

a. $f(x) = x + 7$, $g(x) = x - 7$

b. $f(x) = \frac{1}{2}x + 1$, $g(x) = 2x - 2$



c. $f(x) = \frac{1}{3}x^2$, $g(x) = \sqrt{3x}$

d. $f(x) = \frac{x^5 + 2}{7}$, $g(x) = \sqrt[5]{7x - 2}$

The graph of the inverse function is the reflection of the original function over what line?

VII. Exponential & Logarithmic Functions

Solve each equation.

$y = \log_b x$ if and only if $x = b^y$.

Think of $y = \log_b x$ as the answer to: "To what power must b be raised to obtain x ?"

a. $\left(\frac{1}{3}\right)^x = 27$	b. $5^{3x} = 25^{x-1}$	c. $4^x = 0.25$
d. $10^x = 15$	e. $e^{3x} = 24$	f. $\ln 3x = -0.5003$
g. $\log_x 64 = \frac{1}{2}$	h. $\log_3 x = 5$	i. $\log_4 256 = x$
j. $\log_7(2x+5) = \log_7(x-3)$	k. $\log_2(2x^2) = 5$	l. $\log x = 2.096910013$
m. $256e^{2x} = 1400$	n. $75 = 21(1.05)^t$	o. $10^{x^2+3x-7} = 1,000$

Exponential & Logarithmic Functions (continued)

Write the logs in condensed form.	Write the logs in expanded form.
a. $2\log x - x\log y$	b. $\log x^2 y^3 z^4$
c. $\log x + 2\log y$	d. $\log(x^2 + 1)z$
e. $\log x + \frac{1}{2}\log y - 2\log z$	f. $\log \frac{x^2}{z^6}$
g. $\log x + \log y + \log z - 2\log w$	h. $\log x^2 y$

Exponential & Logarithmic Functions (continued)

Use the equation given and the properties of logs to solve the problems below:	
(1) $A = P \left(1 + \frac{r}{n}\right)^{nt}$ (2) $A = Pe^{rt}$	where: • P = original amount deposited or the initial investment • r = the interest rate expressed as a decimal (5% → 0.05) • n = the number of times a year the interest is paid ("quarterly" → means $n = 4$) • t = the number of years the investment spans
a. Find the value of a \$1,000 investment at 6% interest after 10 years compounded: (a) annually (b) quarterly (c) monthly (d) continuously	b. If you invest \$30,000 at 4.76% interest paid quarterly, how long would it take you to double your money? Round your answer to the nearest hundredth.
c. Suppose \$2,000 is invested in a 3-year certificate of deposit (CD) that earns 6% interest, compounded continuously. How much will the investment be worth after 3 years?	d. You invest \$200 at 12.25% earning continuous interest. How many years does it take for your money to increase 5 times its original value? Round your answer to the nearest tenth.

Student Name _____

P



Algebra II
Test Booklet

Practice Test
Large Print

TEST BOOKLET SECURITY BARCODE

Section 1

(Non-Calculator)

Directions:

Today, you will take Section 1 of the Algebra II Practice Test. You will not be able to use a calculator.

Read each question. Then, follow the directions to answer each question. Circle the answer or answers you have chosen in your test booklet. If you need to change an answer, be sure to erase your first answer completely. If a question asks you to show or explain your work, you must do so to receive full credit. Only responses entered within the space provided will be scored.

If you do not know the answer to a question, you may go on to the next question. If you finish early, you may review your answers and any questions you did not answer in this section ONLY. Do not go past the stop sign.

Directions for Completing the Answer Grids

1. Work the problem and find an answer.
2. Write your answer in the boxes at the top of the grid.
3. Print only one number or symbol in each box. Do not leave a blank box in the middle of an answer.
4. Fractions cannot be entered into an answer grid and will not be scored. Enter fractions as decimals.
5. See below for examples on how to correctly complete an answer grid.

EXAMPLES

To answer -3 in a question, fill in the answer grid as shown below.

-	3					
⊖						
	⊙	⊙	⊙	⊙	⊙	⊙

To answer $.75$ in a question, fill in the answer grid as shown below.

.	7	5				
⊖						
	⊙	⊙	⊙	⊙	⊙	⊙

- 1 Which number is equivalent to $(5 + i)^2$, where i is the imaginary unit?

Select one answer.

- A 24
- B $24 + 10i$
- C 26
- D $26 + 10i$

- 2 What are the solutions of the equation $\frac{x}{x+6} = \frac{2x-3}{x+12}$?

Select **all** that apply.

- A $x = -12$
- B $x = -6$
- C $x = -3$
- D $x = 0$
- E $x = \frac{3}{2}$
- F $x = 3$
- G $x = 6$

- 3 The recursive rule for a sequence is shown, where n represents a positive integer.

$$\begin{aligned} f(1) &= 5 \\ f(n) &= 3 \cdot f(n - 1), \text{ for } n > 1 \end{aligned}$$

Which is an explicit rule for the sequence?

- A** $f(n) = 3(5)^{n-1}$, for $n \geq 1$
- B** $f(n) = 5(3)^{n-1}$, for $n \geq 1$
- C** $f(n) = 3 + 5(n - 1)$, for $n \geq 1$
- D** $f(n) = 5 + 3(n - 1)$, for $n \geq 1$
- 4 The binomial $(x - 2)$ is a factor of the polynomial function $P(x) = x^3 - x^2 + ax - 6$, where a is a constant.

What is the value of a ?

Enter your answer in the space provided.

⊖							
	⊙	⊙	⊙	⊙	⊙	⊙	⊙

- 5 Function f is defined as $f(x) = (0.5)^{2x}$.

Which statement correctly describes how the value of $f(x)$ is changing for every increase of 1 in the value of x ?

Select one answer.

- A** The value of $f(x)$ increases by 25%.
- B** The value of $f(x)$ decreases by 25%.
- C** The value of $f(x)$ increases by 75%.
- D** The value of $f(x)$ decreases by 75%.

- 6 An expression is shown.

$$\left(\frac{64}{27}\right)^{-\frac{1}{3}}$$

Using the properties of exponents, the expression can be rewritten as which of the fractions shown?

A $-\frac{4}{3}$

B $-\frac{3}{4}$

C $\frac{3}{4}$

D $\frac{4}{3}$

- 7 The function f is defined by $f(x) = 2x + 5$.

If $f(a) = c$, which of the following equations gives the value of a in terms of c ?

Select one answer.

A $a = 2c + 5$

B $a = 5c + 2$

C $a = \frac{c - 5}{2}$

D $a = \frac{c}{2} - 5$

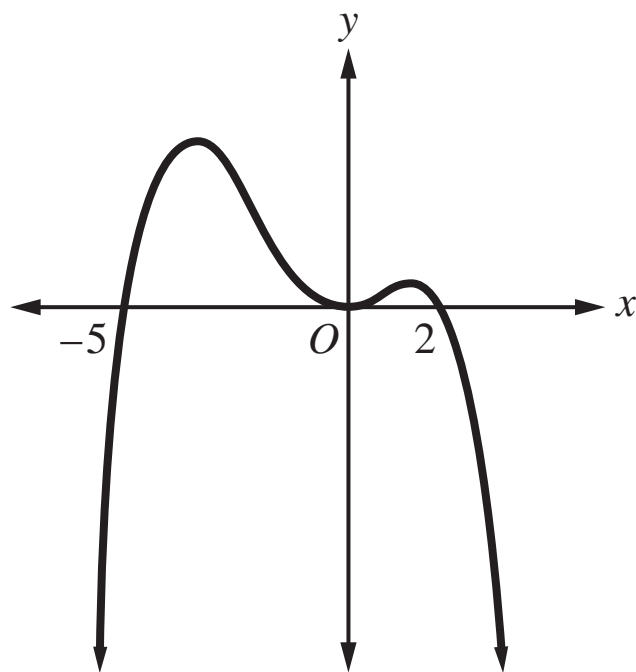
- 8 The expression $x^3 - 3x^2 - 4x + 12$ can be written as $(x^2 - a)(x - b)$, where a and b are constants.

What are the values of a and b ?

- A** $a = 1$ and $b = 12$
- B** $a = 2$ and $b = 6$
- C** $a = 3$ and $b = 4$
- D** $a = 4$ and $b = 3$
- 9 Which value of x makes the equation $5(e)^{4x} = 25$ true?

- A** $x = \ln \frac{5}{4}$
- B** $x = \frac{1}{4} \ln 2$
- C** $x = \frac{1}{4} \ln 5$
- D** $x = \frac{1}{20} \ln 25$

- 10** The graph of a polynomial function is shown in the following xy -plane.

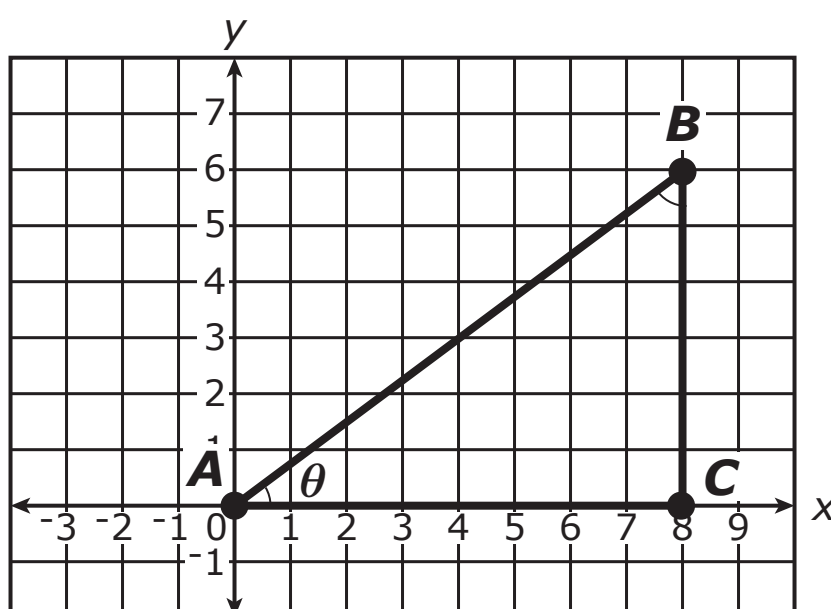


Which equation could represent the graph of the polynomial function?

Select one answer.

- A** $y = x^2(x + 5)(x - 2)$
- B** $y = -x^2(x + 5)(x - 2)$
- C** $y = x^2(x - 5)(x + 2)$
- D** $y = -x^2(x - 5)(x + 2)$

- 11** Right triangle ABC is shown in the xy -coordinate plane. Point B has coordinates $(8, 6)$.



Which statements are true?

Select **all** that apply.

A $\sin \theta = -\frac{4}{5}$

B $\sin \theta = \frac{3}{5}$

C $\sin \theta = \frac{4}{5}$

D $\sin(\pi + \theta) = -\frac{4}{5}$

E $\sin(\pi + \theta) = -\frac{3}{5}$

F $\sin(\pi + \theta) = \frac{3}{5}$

12 Which table shows the three quantities 4^7 , $(4^6)(4^2)$, and $9^{3.5}$ in order from **least** to **greatest**?

A

Least	$9^{3.5}$
	$(4^6)(4^2)$
Greatest	4^7

B

Least	$9^{3.5}$
	4^7
Greatest	$(4^6)(4^2)$

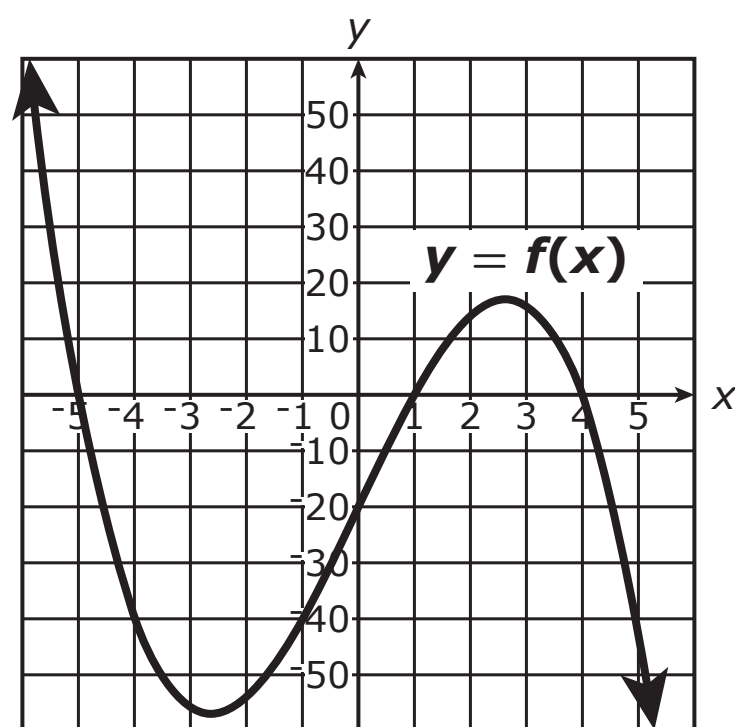
C

Least	4^7
	$9^{3.5}$
Greatest	$(4^6)(4^2)$

D

Least	4^7
	$(4^6)(4^2)$
Greatest	$9^{3.5}$

- 13** The graph of the polynomial function f is shown in the xy -coordinate plane.



Which of the equations shown could define f ?

- A** $f(x) = -(x + 5)(x - 1)(x - 4)$
- B** $f(x) = (x + 5)(x - 1)(x - 4)$
- C** $f(x) = -(x - 5)(x + 1)(x + 4)$
- D** $f(x) = (x - 5)(x + 1)(x + 4)$



You have come to the end of Section 1 of the test. Review your answers from Section 1 only.





GO ON TO NEXT PAGE



Section 2

(Calculator)

Directions:

Today, you will take Section 2 of the Algebra II Practice Test. You will be able to use a calculator.

Read each question. Then, follow the directions to answer each question. Circle the answer or answers you have chosen in your test booklet. If you need to change an answer, be sure to erase your first answer completely. If a question asks you to show or explain your work, you must do so to receive full credit. Only responses entered within the space provided will be scored.

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-	3					
⊖						
	⊙	⊙	⊙	⊙	⊙	⊙

To answer $.75$ in a question, fill in the answer grid as shown below.

.	7	5				
⊖						
	⊙	⊙	⊙	⊙	⊙	⊙



Mathematics

- 1 The table shows the percent of US households with computers for selected years between 1984 and 2003.

Percent of Households with Computers

Year	Percent
1984	8.2
1989	15.0
1993	22.9
1997	36.6
2000	51.0
2001	56.3
2003	61.8

Which of the equations shown **best** models the data in the table, where y is the percent of households with computers and x is the number of years after 1980?

A $y = 5.6(1.08)^x$

B $y = 5.6(1.09)^x$

C $y = 5.6(1.11)^x$

D $y = 5.6(1.14)^x$



2 The functions f and g are defined as shown.

$$f(x) = x^3 + x^2 - 2x$$

$$g(x) = 0.5x^2 + 1$$

The values of x for which $f(x) = g(x)$ lie in which of the intervals shown?

Select **all** that apply.

A $-3 < x < -2$

B $-2 < x \leq -1$

C $-1 < x \leq 0$

D $0 < x \leq 1$

E $1 < x \leq 2$

F $2 < x < 3$



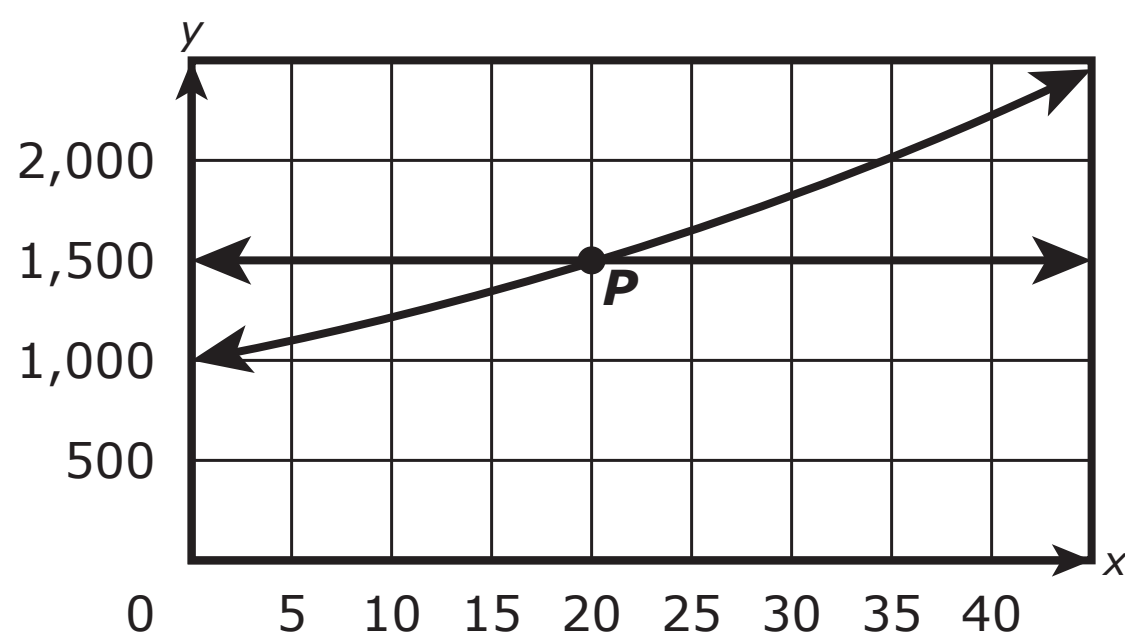
Mathematics

Section 2

- 3** An account that pays 0.5% interest was opened with an initial deposit, and no additional deposits to or withdrawals from the account were made. The equation $y = 1000(1.005)^{4x}$ models the amount, in dollars, in the account x years after it was opened.

The graph of the model and the graph of the equation $y = 1500$ are shown in the xy -coordinate plane.

The intersection of the two graphs is marked with the point P .



What is revealed by the coordinates of the point P ?

- A** the yearly interest rate for the account
- B** the amount of the initial deposit in the account
- C** the number of years it takes for the account value to reach \$1500
- D** the number of years it takes for the initial amount in the account to double



GO ON TO NEXT PAGE



Mathematics

- 4 Claim: There is no quadratic equation with real coefficients for which $x = -5i$ is a solution.

Part A

Show that the claim is not correct by providing an example of a quadratic equation for which $x = -5i$ is a solution. Include any other solutions to your equation in your answer.

Enter your answer and your explanation in the space provided.

4 Part A

**Part B**

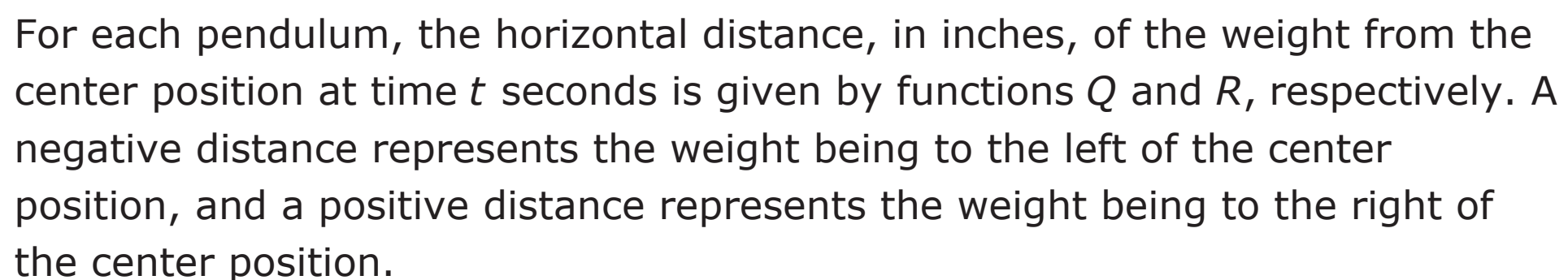
Is it possible for a quadratic equation with real coefficients to have $x = -5i$ as its **only** solution? Justify your answer.

Enter your answer and your justification in the space provided.

4 Part B



5 For the two pendulums shown, a weight swings back and forth from left to right. Both pendulums start at the left position at time $t = 0$ seconds.



$$\begin{aligned} Q(t) &= -6\cos(\pi t) \\ R(t) &= -4\cos\left(\frac{4}{3}\pi t\right) \end{aligned}$$

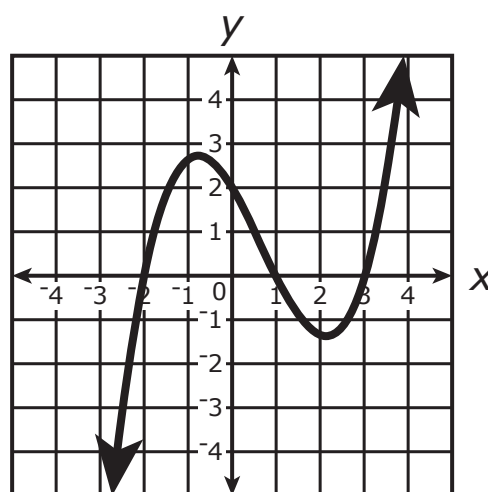
The maximum horizontal distance of pendulum Q to the right of its starting position is how much greater, in inches, than the maximum horizontal distance of pendulum R to the right of its starting position?

Enter your answer in the space provided.

⊖						
	⊙	⊙	⊙	⊙	⊙	⊙



- 6 Consider the polynomial function P , defined by $P(x) = x^3 + cx^2 + x + 2$, where c is a constant, and consider the graph shown in the xy -coordinate plane.



If $P(-2) = 0$, can the graph shown be the graph of $y = P(x)$?

- A** Yes, because c would equal -2 , and the points $(1, 0)$ and $(3, 0)$ would be on the graph of $y = P(x)$.
- B** Yes, because c would equal 2 , and the points $(1, 0)$ and $(3, 0)$ would be on the graph of $y = P(x)$.
- C** No, because c would equal -2 , and the points $(-2, 0)$ and $(3, 0)$ would not be on the graph of $y = P(x)$.
- D** No, because c would equal 2 , and the points $(1, 0)$ and $(3, 0)$ would not be on the graph of $y = P(x)$.



Mathematics

Section 2

- 7** When a customer pays with a debit card at a clothing store, the owner pays a processing fee. The function P represents the processing fee, in dollars, for a debit card purchase of x dollars.

$$P(x) = 0.035x + 0.15$$

How is the processing fee calculated based on the purchase amount?

- A** The processing fee consists of a fixed amount of \$0.15 plus 0.35% of the purchase amount.
- B** The processing fee consists of a fixed amount of \$0.15 plus 3.5% of the purchase amount.
- C** The processing fee consists of a fixed amount of \$0.35 plus 1.5% of the purchase amount.
- D** The processing fee consists of a fixed amount of \$0.35 plus 15% of the purchase amount.

- 8** An equation is shown.

$$\sqrt{x^2 + 19} - 100 = 0$$

What are the solutions of the equation?

- A** $x = 9$ only
- B** $x = 3\sqrt{1109}$ only
- C** $x = -9$ and $x = 9$
- D** $x = -3\sqrt{1109}$ and $x = 3\sqrt{1109}$





You have come to the end of Section 2 of the test. Review your answers from Section 2 only.





Section 3

(Calculator)

Directions:

Today, you will take Section 3 of the Algebra II Practice Test. You will be able to use a calculator.

Read each question. Then, follow the directions to answer each question. Circle the answer or answers you have chosen in your test booklet. If you need to change an answer, be sure to erase your first answer completely. If a question asks you to show or explain your work, you must do so to receive full credit. Only responses entered within the space provided will be scored.

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Directions for Completing the Answer Grids

1. Work the problem and find an answer.
2. Write your answer in the boxes at the top of the grid.
3. Print only one number or symbol in each box. Do not leave a blank box in the middle of an answer.
4. Fractions cannot be entered into an answer grid and will not be scored. Enter fractions as decimals.
5. See below for examples on how to correctly complete an answer grid.

EXAMPLES

To answer -3 in a question, fill in the answer grid as shown below.

-	3					
⊖						
	⊙	⊙	⊙	⊙	⊙	⊙

To answer $.75$ in a question, fill in the answer grid as shown below.

.	7	5				
⊖						
	⊙	⊙	⊙	⊙	⊙	⊙



Mathematics

- 1 The function C approximates the number of calories recommended per day for a dog, based on the weight, in pounds, of the dog.

The table gives values of C for selected values of p , where $C(p)$ represents the recommended number of calories per day for a dog that weighs p pounds.

Recommended Number of Calories for Dogs

p (pounds)	$C(p)$
11	250
22	450
33	750
44	1000
55	1250
66	1500
77	1700
88	1880

Which of the equations shown **most** closely approximates C ?

- A** $C(x) = 18p + 22$
- B** $C(x) = 22p + 18$
- C** $C(x) = 40(1.06)^p$
- D** $C(x) = 48(1.07)^p$



2 Claim: $\sqrt[3]{x} \leq x$, where x is a real number

For which intervals is the claim true for all values of x in the interval?

Select **all** that apply.

A $x > 1$

B $0 < x < 1$

C $-1 < x < 0$

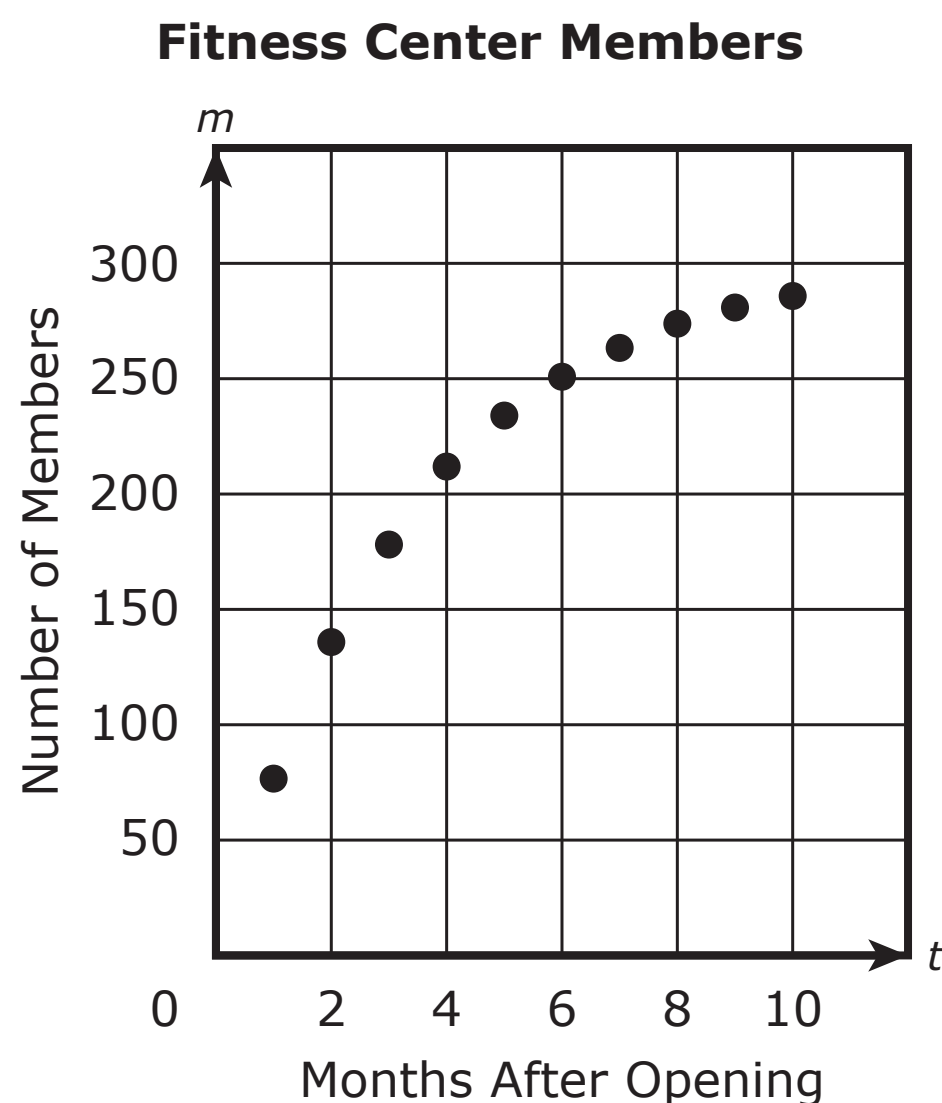
D $-2 < x < -1$

E $x < -2$



Mathematics

- 3 The number of members, m , that a new fitness center expects to have t months after the opening of the center is shown in the scatter plot.



The fitness center constructed the model $m(t) = 300 - 280(0.76)^t$ to project the number of members that can be accommodated t months after opening.

- What is the average rate of change, in members per month, in the number of members the fitness center expects to have from the 4th to the 8th month after opening? Show your work.
- What is the significance of the number 300 in the model? Explain your answer.

Enter your answers and your work and explanation in the space provided.



3

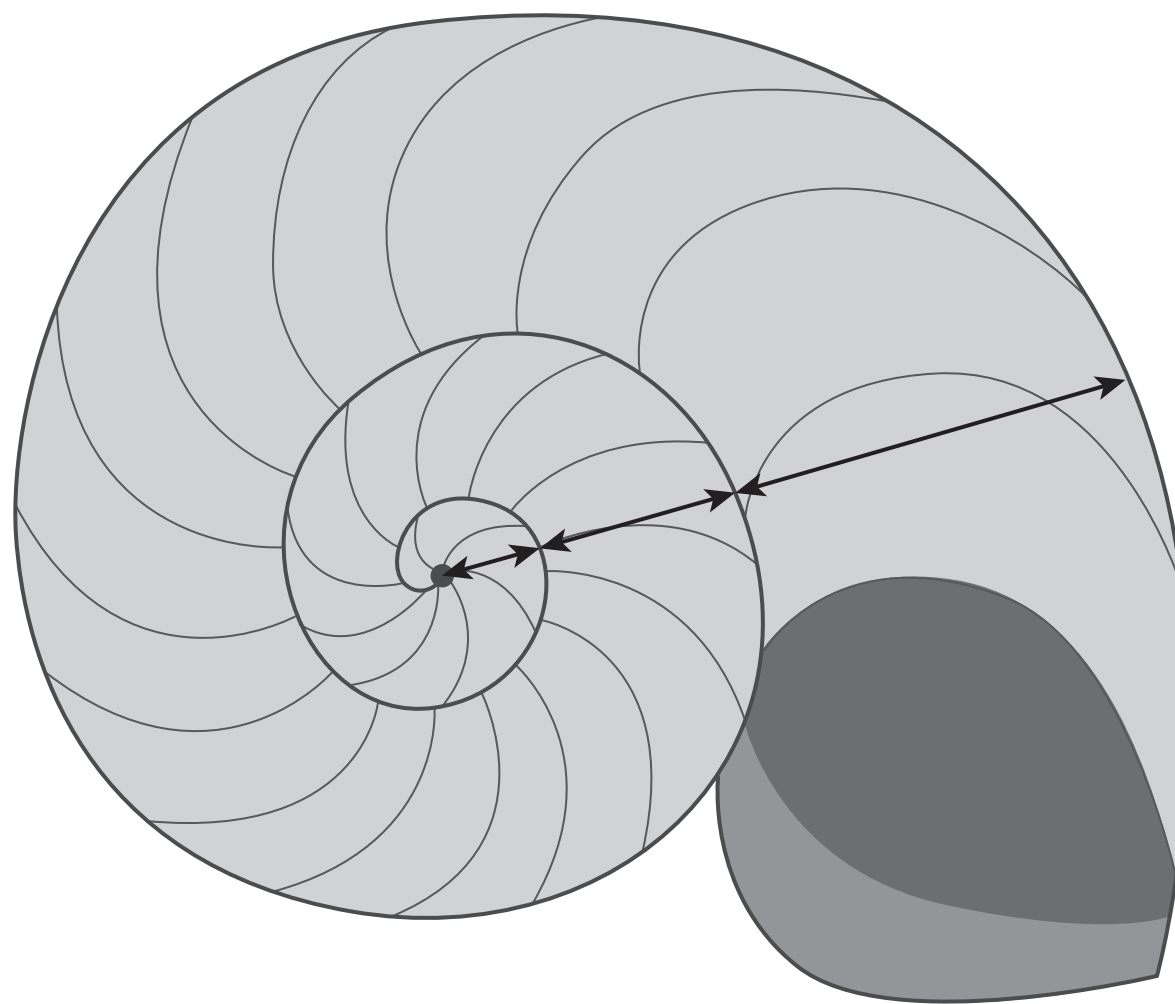
Section 3



Mathematics

- 4 The students in a class are analyzing several examples of spirals in nature.

The students measured the distances shown in the picture for different shells.



Section 3

Each student measured the three distances shown in the picture by starting from the outside of the shell and moving toward the center.

One student's measurements were 8.1 centimeters, 4.2 centimeters, and 2.1 centimeters. A second student's measurements of a different shell were 7.6 centimeters, 3.7 centimeters, and 1.8 centimeters.

Which equation could be used to model the students' data, where a represents the first measurement, in centimeters, and d represents the n th measurement, in centimeters, where $n = 1, 2$, or 3 ?

A $d = a\left(\frac{1}{2}\right)^{n-1}$

B $d = a(2^{n-1})$

C $d = a\left(\frac{1}{2}\right)^n$

D $d = a(2^n)$



GO ON TO NEXT PAGE

Section 3



Mathematics

- 5 Consider the functions P and Q , defined as shown.

$$P(x) = x^2 + 7x - 14$$

$$Q(x) = -3x + 10$$

In the xy -coordinate plane, what are the coordinates of the points at which the graphs of the equations $y = P(x)$ and $y = Q(x)$ intersect?

Explain how you determined your answer.

Enter your answer and your explanation in the space provided.



5



Mathematics

- 6** The approximate number of cars parked in a garage x hours after 6:00 a.m. can be modeled by the function $y = 180 + 130\sin\left(\frac{x}{12}\pi\right)$, where $0 \leq x \leq 16$.

Based on the model, for which time interval did the number of cars parked in the garage increase?

- A** 7:00 a.m. to 9:00 a.m.
- B** 1:00 p.m. to 4:00 p.m.
- C** 5:00 p.m. to 7:00 p.m.
- D** 8:00 p.m. to 10:00 p.m.





Section 3

You have come to the end of Section 3 of the test. Review your answers from Section 3 only.





Section 4

(Calculator)

Directions:

Today, you will take Section 4 of the Algebra II Practice Test. You will be able to use a calculator.

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EXAMPLES

To answer -3 in a question, fill in the answer grid as shown below.

-	3					
⊖						
	⊙	⊙	⊙	⊙	⊙	⊙

To answer $.75$ in a question, fill in the answer grid as shown below.

.	7	5				
⊖						
	⊙	⊙	⊙	⊙	⊙	⊙



Mathematics

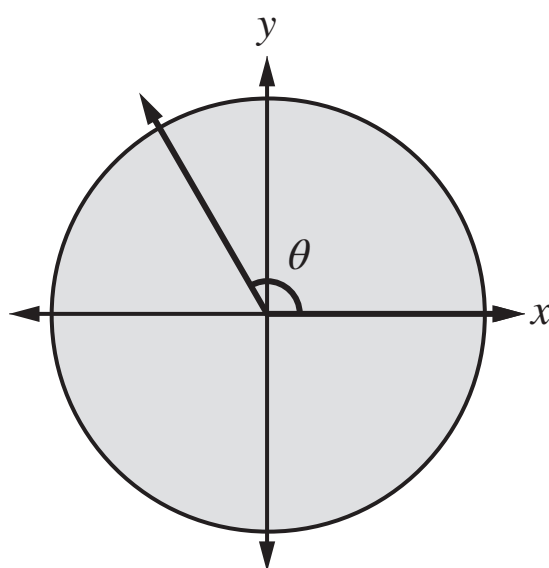
- 1 The half-life of the isotope Francium-223 is 22 minutes. For a sample that starts with F grams of Francium-223, the expression $F(0.5)^{\frac{m}{22}}$ represents the number of grams remaining after m minutes.

Which expression represents the number of grams of Francium-223 remaining after h hours?

- A $F(0.151)^h$
- B $F(0.5)^{\frac{h}{60}}$
- C $F(0.5)^{60h}$
- D $F(0.969)^h$



- 2 Angle θ is positioned in a circle with its initial ray on the positive x -axis and its terminal ray $\frac{1}{3}$ of one complete revolution about the origin, as shown in the following diagram.



Which of the following statements correctly expresses the measure of θ in both degrees and radians?

Select one answer.

- A** The measure of angle θ is 120 degrees or $\frac{\pi}{3}$ radians.
- B** The measure of angle θ is 120 degrees or $\frac{2\pi}{3}$ radians.
- C** The measure of angle θ is 150 degrees or $\frac{5\pi}{12}$ radians.
- D** The measure of angle θ is 150 degrees or $\frac{7\pi}{12}$ radians.
- 3 Claim: For every value of the constant c , the equation $x^4 + (1 - c^2)x^2 - c^2 = 0$ has at least two distinct real solutions.

What value of c refutes the claim?

Enter your answer in the space provided.

⊖							
⊙	⊙	⊙	⊙	⊙	⊙	⊙	⊙



- 4 The table shows the box office revenue, in millions of dollars, for the first 10 weeks after a movie opened in theaters.

Box Office Revenue for a Movie

Number of Weeks after Opening	Box Office Revenue (in millions)
0	\$85.63
1	\$51.97
2	\$37.21
3	\$25.20
4	\$15.22
5	\$11.17
6	\$7.02
7	\$4.03
8	\$4.42
9	\$1.95

**Part A**

Create a function to model the data. Explain why you chose the type of function and explain the meaning of the parameters of the function.

Enter your answer and your explanation in the space provided.

4 Part A



Mathematics

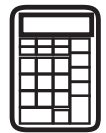
Part B

The theaters will stop showing the movie when the weekly box office revenue is below \$100,000. In what week does your model predict that the theaters will stop showing the movie?

Show your work or explain how you found your answer.

Enter your answer and your work or explanation in the space provided.

4 Part B



- 5** Consider the equation $a = \sqrt{x + b}$, where a and b represent real numbers.

Which statement is true about the number of real solutions to the equation?

- A** The number of real solutions depends only on the value of a .
- B** The number of real solutions depends only on the value of b .
- C** The number of real solutions depends on the relationship between a and b .
- D** The values of a and b do not affect the number of real solutions.

- 6** The world population in 1997 was 5.88 billion.

The world population in 2017 was 7.53 billion.

Assume that the ratio between the population in two consecutive years was constant between 1997 and 2017.

Which equation can be used to find r , the rate of growth per year of the world population?

- A** $5.88 = 7.53r^{10}$
- B** $5.88 = 7.53r^{20}$
- C** $7.53 = 5.88r^{10}$
- D** $7.53 = 5.88r^{20}$



Mathematics

7 Consider the system of equations shown.

$$y = (x - 3)^2$$
$$y = x - 1$$

On the coordinate plane, what are the coordinates of the points that represent solutions (x, y) of the system?

A $(-5, -6)$ and $(-2, -3)$

B $(-5, -6)$ and $(2, 1)$

C $(5, 4)$ and $(-2, -3)$

D $(5, 4)$ and $(2, 1)$

8 A kitchen supply store sells a set of 4 food storage containers.

- From largest to smallest, the volumes of the containers are $v_1, v_2, v_3,$ and v_4 .
- v_n is 20% less than v_{n-1} , where n is 2, 3, or 4.
- $v_n = kv_{n-1}$, where n is 2, 3, or 4.

What is the value of k ?

A $\frac{3}{4}$

B $\frac{4}{5}$

C $\frac{5}{4}$

D $\frac{4}{3}$





You have come to the end of Section 4 of the test. Review your answers from Section 4 only.

Section 4



P

ALG II

Semester Exam - Algebra II

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Simplify each expression.**Unit 1**

1) $-4x(x + 4) + x(x + 4)$

2) $-6(2 - 6x) - 5(x - 4)$

Solve each equation.

3) $\frac{p}{4} - \frac{p}{5} = -50p$

4) $4|3b + 5| - 2 = -2$

5) $\frac{1}{6}p + \frac{19}{6} = -\frac{1}{6}$

6) Solve the equation $r = \frac{2t}{g}$ for g.

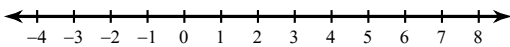
Solve each equation. Write non-integer answers in fractional form.

7) $|-5p| - 1 = 24$

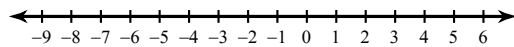
8) $1 - 6|5m - 5| = 31$

Solve each compound inequality and graph its solution.

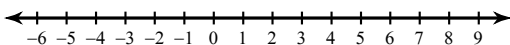
9) $-2 < 6m + 10 < 34$



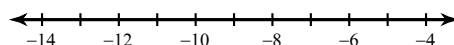
10) $-5 + 8r < -37$ or $9r + 8 > 17$

**Solve each inequality and graph its solution.**

11) $-22 \leq 2 - 3m \leq 14$



12) $-5 - r > 7$



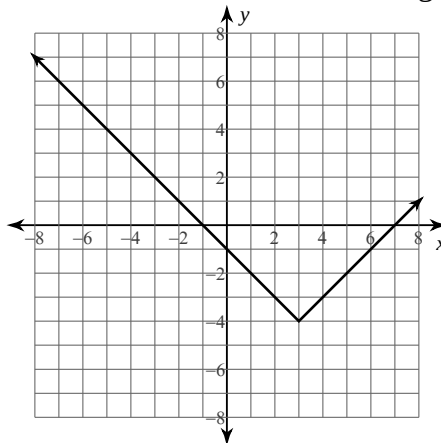
Solve for the indicated variable in parenthesis.

13) $\frac{a+b}{k} = 1$ (b)

14) $4x - 5y = 9$ (y)

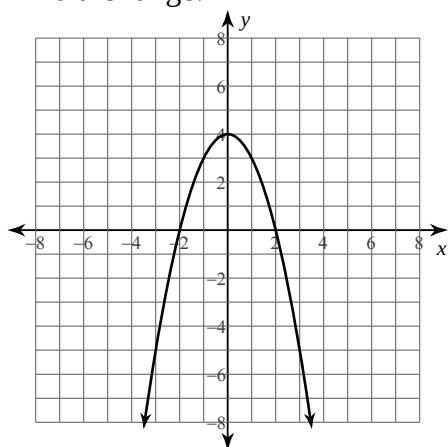
15) $4x - 5y = 9$ (y)

16) Find the domain of the following graph.



Unit 2

17) Find the range.



18) Habitat for Humanity builds 5 houses in 6 weeks and then manages to build 15 houses in 14 weeks. What's the average rate of change of houses built per week?

Find the maximum or minimum of the following function.

19) $f(x) = -2(x+9)^2 + 20$

Solve.

21) $2(x-13)^2 - 25 = 175$

Evaluate each function.

23) $h(x) = 2x^2 - 6x$; Find $h(-5)$

Put the following into standard form.

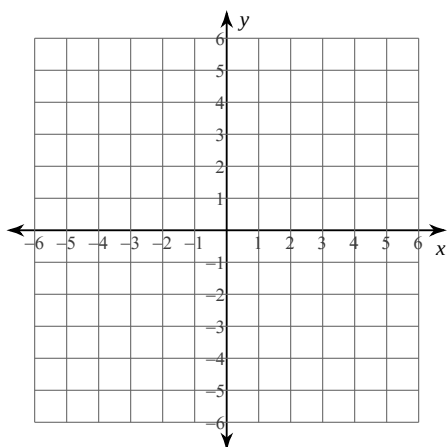
20) $f(x) = -(x+4)^2 - 12$

Evaluate or solve each function.

22) $k(t) = -6t + 14$; Find $k(-5)$

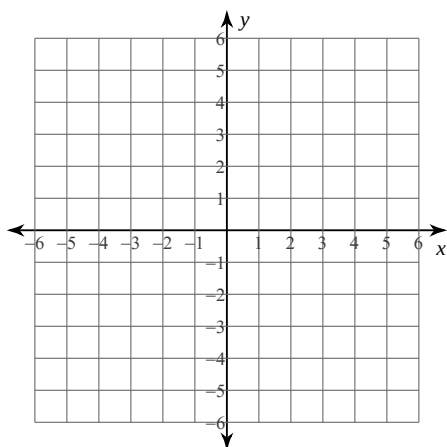
Sketch the graph of each line.

24) $x - 5y = -15$



Find the x- and y-intercepts, then graph.

25) $3x - 4y = -12$



Write the slope-intercept form of the equation of the line described.

26) through: $(3, -3)$, perp. to $y = \frac{3}{8}x + 3$

27) through: $(3, 3)$, parallel to $y = \frac{4}{3}x - 2$

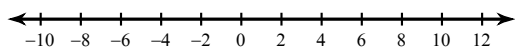
Write the slope-intercept form of the equation for each.

28) through: $(-4, -2)$ and $(-5, 2)$

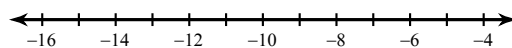
Unit 3

Solve each inequality and graph its solution.

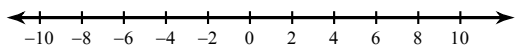
29) $|6a| + 10 \geq 52$



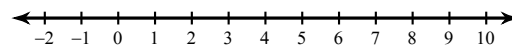
30) $-9|x + 10| - 10 > -46$



31) $2|-7 + 7a| - 6 \geq 78$

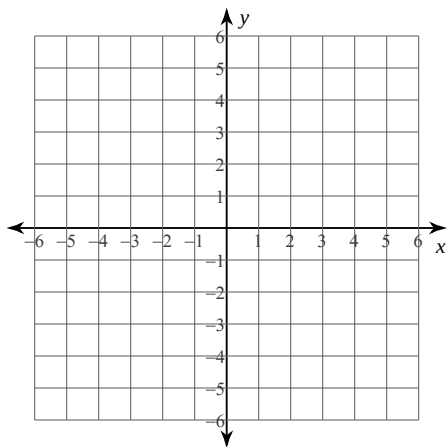


32) $10|10n + 7| - 8 < 62$

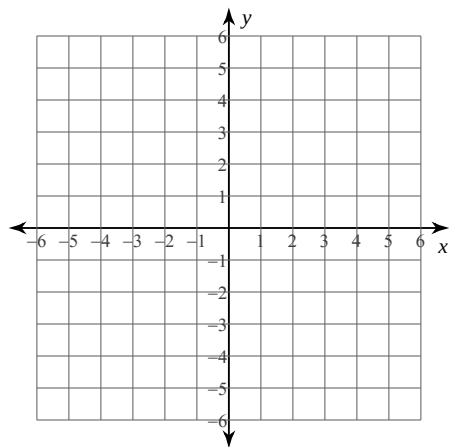


Graph each equation.

33) $y = -2|x| + 3$



34) $y = 3|x - 2| + 2$

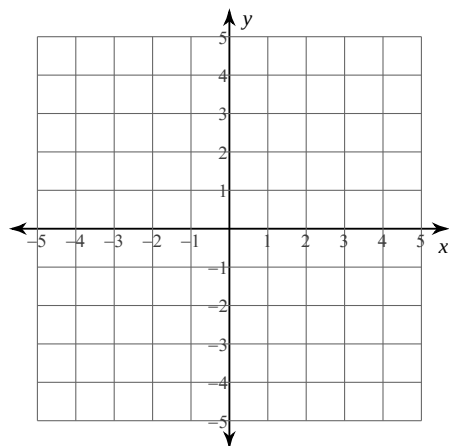


Solve using any method. Unit 4

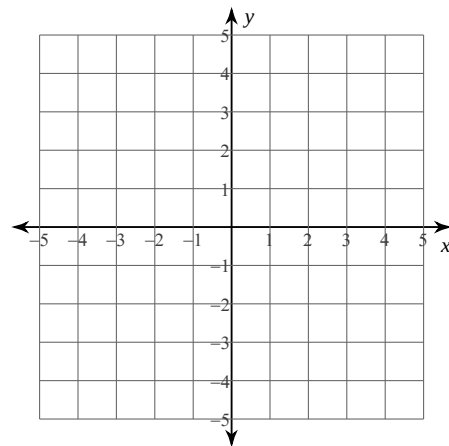
35) $1 - y = -x$
 $0 = -x + y - 1$

36) $7x + y = -4$
 $-2 = -y - x$

37) $-3y - 6 = -2x$
 $x = -1 + \frac{1}{2}y$

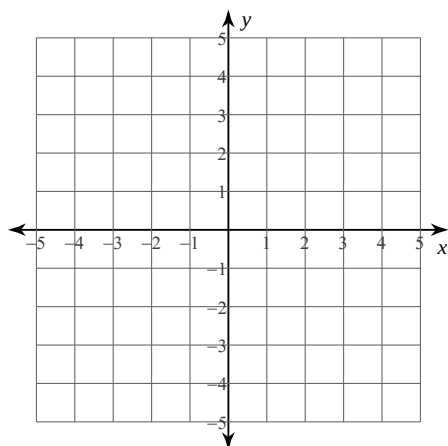


38) $y \leq -x - 2$
 $y < 2x + 1$



39) $y < \frac{2}{3}x - 3$

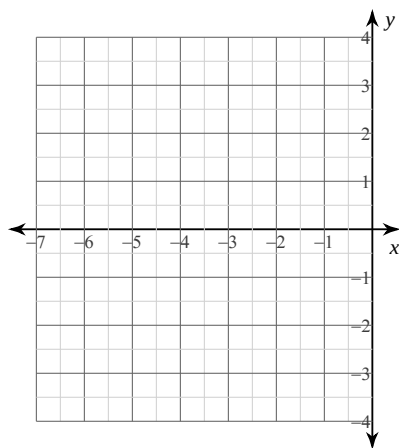
$y \geq -\frac{4}{3}x + 3$



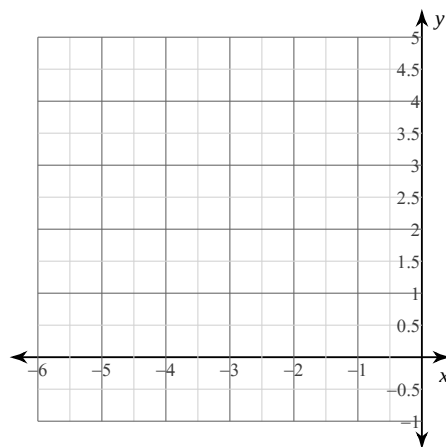
Sketch the graph of each function.

Unit 5

40) $y = -(x + 4)^2 + 2$



41) $y = -x^2 - 6x - 5$



Factor each completely.

42) $3b^4 + 3b^3 - 36b^2$

43) $12x^3 + 14x^2 - 48x$

44) $v^2 - 6v + 9$

45) $12m^2 - 27$

Simplify.

46) $\frac{2}{2 + 5\sqrt{2}}$

Unit 6 47) $(2 + 2i) + (-4 + 4i)$

48) $(7 - 6i) - (-5 + 2i)$

49) $(-7 - 8i)(-5 + 8i)$

50) $(5 - 5i)^2$

51) $\frac{2i}{7+9i}$

52) $\frac{5-3i}{-4-i}$

Solve each equation by completing the square.

53) $x^2 - 16x - 70 = -4$

54) $9p^2 + 56 = 18p$

Solve each equation with the quadratic formula.

55) $4b^2 + 4b = -4$

56) $10v^2 - 3v - 9 = -5 + 5v^2$

Find the discriminant of each quadratic equation then state the number and type of solutions.

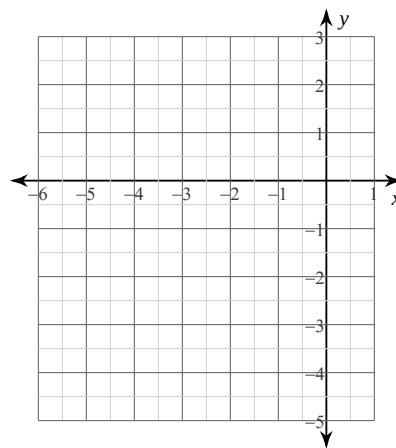
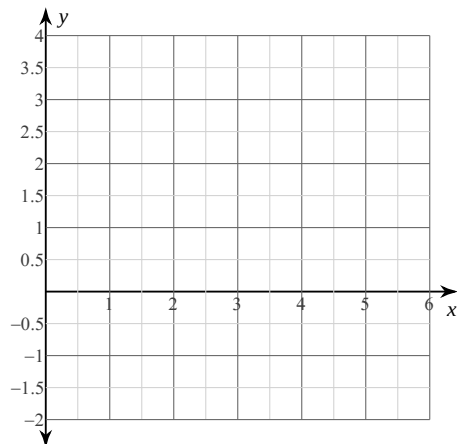
57) $-3x^2 + 5x - 2 = -10$

58) $6n^2 - n - 9 = -9$

Write in vertex form. Find the roots. Graph and label vertex and roots.

59) $y = -x^2 + 4x - 1$

60) $y = \frac{1}{2}x^2 + 4x + 6$



Application and Extension

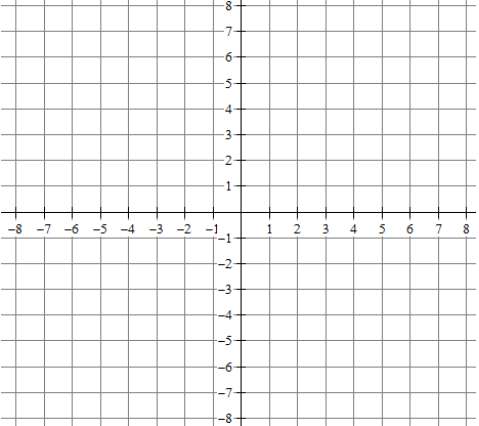
UNIT 1: Algebro's shirts are all the rage! To make an Algebro new shirt, a company charges \$60 for a design set-up (one time fee), and then \$22 per shirt. You plan on ordering several shirts for your Algebro fan-club.

- a. Write an expression that models the total cost for the number of shirts you order including the design-set-up fee.
- b. Use your expression to find the total cost of ordering 20 shirts for your fan-club.

UNIT 2: Timmykat's BARBIE FANPAGE website is BOOMING! A big company offers to come in and pay him to advertise on their page because he is getting so many hits. The CEO of the company says that at 10 hits he'll get \$4 and that when he gets to 100 hits he'll get \$22.

- a) What is the average rate of change of dollars per hit that the CEO is offering?
- b) What's the equation of the line for this situation?
- c) If Timmykat had 1000 hits in one day how much money would the company give him?
- d) How many hits would it take for Timmykat to make \$1000 from the company?

UNIT 3: Given the piecewise function

$f(x) = \begin{cases} -\frac{1}{2}x + 3 & x < 0 \\ 3x - 1 & x \geq 0 \end{cases}$	Find... a. $f(4) =$ b. $f(-5) =$ c. $f(0) =$	Graph $f(x)$ 
---	--	--

UNIT 4: Sully loves to travel in style... so he invests in an Alge-Blimp! Sully charges tickets to ride in his Alge-Blimp as either an "Adult" or "Child" ticket. Bean shows up with 7 adults and 7 children and it costs him \$196. He had so much fun, he took his whole 3rd period class (14 children and 1 adult) to travel on the Alge-Blimp again and paid \$210 (*true story... no hot air here!*). Find the cost of 1 adult and 1 child ticket.

UNIT 5: Bean realizes that he's going to need more room at his house for his ever expanding family. He decides to give his kids more room in their play room. Currently the room is 15 feet by 20 feet and he decides that he'll add 450 square feet to that by adding the same distance, x , to both the length and width of the room.

a) Find the current area of the room.

b) Write an equation that models adding 450 square feet to the area by adding the same distance x to both the length and the width.

c) Solve the equation and find the new dimensions of the room that Bean is going to build.

UNIT 6: An object that travels up and down is modeled by the quadratic function below:

$$h(t) = -16t^2 + v_0t + h_o$$

h = height of object (feet), t = time (seconds), v_0 = initial velocity of object (ft/sec), h_o = initial height of object (feet)

Bob fires a gun straight up from 7 feet. The bullet is shot straight into air with a velocity of 240 ft/sec.

a. Write the equation that models this.

b. What does $h(3)$ mean? Find it.

c. Find the maximum height of the bullet.

d. When does the rocket hit the ground?

**Algebra II R Semester 1 Final Exam Review
2012**

Name _____

1. Multiply: $(2x^2 - 5)(3x^2 + 7x)$

2. Evaluate: $a = 2$, $b = -3$ and $c = 4$

$$\frac{a(b-c)}{c(a+b)} =$$

3. Solve for x :

$$3 + 8(5x - 2) = 2x - (7x - 4)$$

4. Solve the equations for “ a ”

$$K = \frac{1}{2}ap \qquad H = \frac{1}{3}at$$

5. The Perimeter of a rectangle is 12. The length is 2 more than the width. Find both dimensions.

6. Solve the inequality, and graph the solution on a number line

$$3x - 5 \leq 9x - 8$$



7. **Solve for x :**

$$|x - 3| + 5 = 9$$

8. **Factor:**

$$20x^5y^6z^7 + 15x^2y^4z^3 - 35x^7y^2z^2$$

9.

a. State the slope of the line containing the points $(4, -3)$ and $(-9, 6)$. _____

b. What is the slope of a line parallel to this line? _____

c. What is the slope of a line perpendicular to this line? _____

10. **Factor:**

a. $9x^2 + 12x + 4$ _____

b. $16x^2 - 8x + 1$ _____

11. Describe these lines as parallel, intersecting, or same. *hint: check the slopes*

a. $y = \frac{3}{4}x - 1$

$y = \frac{3}{4}x + 5$ _____

b. $y = -\frac{4}{3}x + 2$

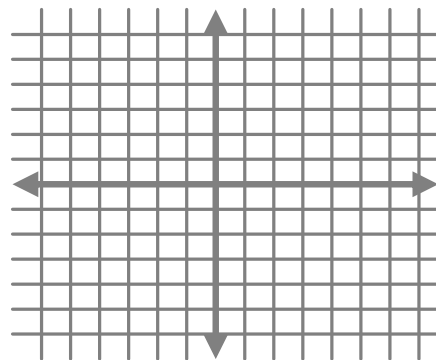
$y = \frac{4}{3}x - 2$ _____

c. $x + y = 4$

$2x + 2y = 8$ _____

12. Graph: $2x - 3y = 6$

$m =$ _____ $b =$ _____



13. Write the equation of a line in slope-intercept form. *hint: for problem b use point-slope*

$$y - y_1 = m(x - x_1)$$

a. $m = 6$ $b = 4$ _____

b. $m = -5$ *pt.* $(-2, 1)$ _____

14. Write an equation of a line in Standard form. *hint: for problem b use point-slope*

a. $m = -4$ $b = 6$ _____

b. $m = 3$ *pt.* $(1, -5)$ _____

15. $f(x) = 2x^2 + 3x - 6$

$f(1) =$ _____

$f(-3) =$ _____

16. Which of the following are functions?

a. $\{(4, 5), (4, 2), (5, 2)\}$ _____

b. $\{(5, -4), (6, -4), (3, 1)\}$ _____

17. Factor

a. $x^3 + 125$ _____

b. $8y^3 - 1$ _____

18. Solve the system:

$$2x + 3y = 7$$

$$5x - 2y = -30$$

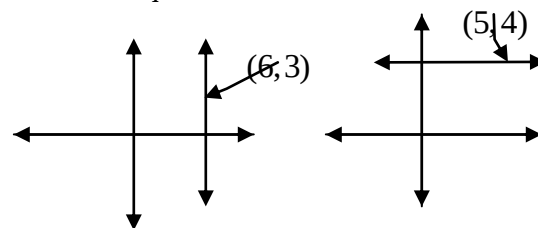
19. Solve the system
 $y = 2x + 1$ $4x + 3y = 23$

20. Factor:
 a. $2x^2 - 32$ _____

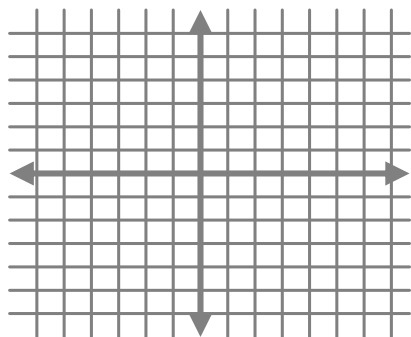
b. $xy^2 - 49x$ _____

21. A collection of dimes and quarters has \$1.55. There is a total of 8 coins. Write and solve a system of equations to find the number of each.

22. State the equation of each line.



23. Graph and shade the system
 $y \geq -2$
 $y < \frac{2}{5}x + 2$



24. Name all of the subsets of the real numbers to which each is a member.

Use: R I Q Z W N

$\sqrt{2}$ _____

-2.5 _____

5 _____

25. Factor:
 $x^2 + 2x - 3 =$ _____

$5x^2 - 12x + 4 =$ _____

26. Simplify:
 $(3x - 2)(4x + 5) =$ _____

$(4x - 3y)(4x + 3y) =$ _____

27. $(2x^3 + 3x^2 + 2) + (6x^2 + 4x - 5)$

$(2x^3 + 3x^2 + 2) - (6x^2 + 4x - 5)$

28. Solve the system: (____, ____, ____)

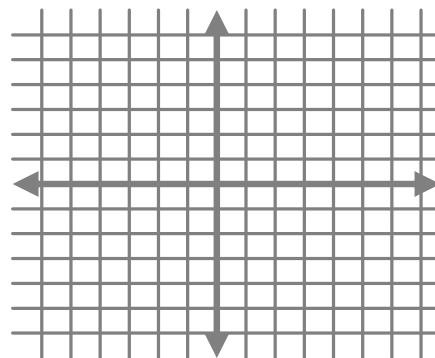
$$\begin{aligned} x + y + z &= 2 \\ 2x + y - z &= -1 \\ x + 2z &= 5 \end{aligned}$$

29 Simplify:

$2x(4x^2 - 3) = \underline{\hspace{2cm}}$

$3x^2(5x + 1) = \underline{\hspace{2cm}}$

30. Graph. $y = |x + 2| - 3$



31. Multiply:

$(6x - 1)^2 = \underline{\hspace{2cm}}$

$(4x + 7)^2 = \underline{\hspace{2cm}}$

32. Multiply:

$(2x - 1)(x^2 - 6x + 3) = \underline{\hspace{2cm}}$

33. Simplify:

$$\frac{4x^2y^4 - 2xy^3}{3xy^6}$$

$$\frac{5xy^2 - 10x^2y^6}{15x^3y^4}$$

34. Divide: 1st by long

$$3x - 2 \overline{) 6x^2 - 10x + 4}$$

2nd by synthetic

$$\frac{8x^3 - 6x^2 + x - 3}{x + 1}$$

35. Solve and graph the Solution Set.

$$4x - 3 > 9 \text{ or } 4x - 3 < -11$$



36. Solve and graph the solution set.

$$|3x - 4| \leq 17$$



37. Solve by any method

$$4x - 10y = -3$$

$$2x + y = 5$$

$$x = \underline{\hspace{2cm}}$$

$$y = \underline{\hspace{2cm}}$$

38. Factor each expression.

$$12x^2 - 71x - 6$$

$$x^4 - 13x^2 + 36$$

$$4x^3 - 100x$$

$$x^3 - 125$$

$$27x^3 + 64y^3$$

$$2x^2 + 7x - 15$$

39. $(x - 3)^5$

$$(2x - b)^4$$

40. Factor by grouping

$$cx - cy + dx - dy \quad \underline{\hspace{2cm}}$$

$$ax^2 + bx^2 - 4a - 4b \quad \underline{\hspace{2cm}}$$

41. $i^{17} = \underline{\hspace{2cm}}$
 $i^{27} = \underline{\hspace{2cm}}$
 $i^{36} = \underline{\hspace{2cm}}$

42. $(3 - 2i)(5 + 7i)$

$(4 - 2i)(4 + 2i)$

43. $\frac{\sqrt{3}}{4i}$

44. $x^2 + 8 = 3$

$\frac{3}{6 - 2i}$

46. Graph $y = 2x^2 + 4x - 7$

45. $\frac{3 - 7i}{5 + 4i}$

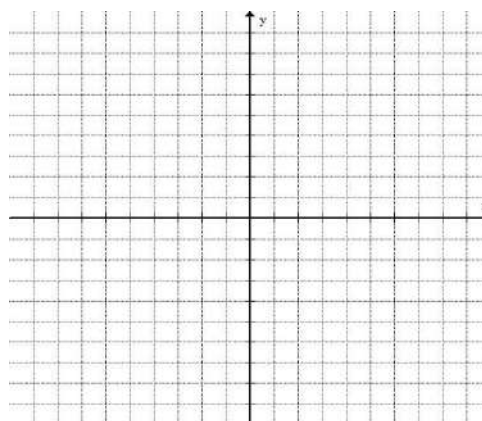
Which direction does the graph open? $\underline{\hspace{2cm}}$

Find the vertex $V(\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$

Write the equation for the axis of symmetry.

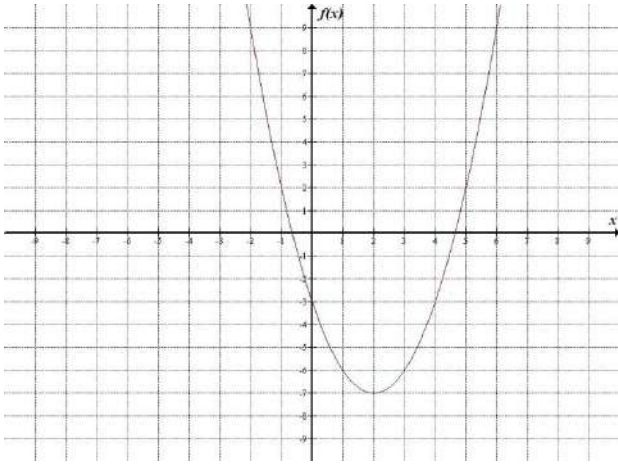
Find the y-int. $(\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$

47. $9x^2 + 7 = 4$



48. Identify the solutions to the following quadratic equation by using the given graph.

$$x^2 - 4x - 3 = 0$$



$$\underline{\quad} < x < \underline{\quad} \quad \text{and} \quad \underline{\quad} < x < \underline{\quad}$$

49. Write a quadratic equation in standard form with solutions, $x = -3$ and $x = 4$. Use integers for a , b and c .

50. Write a quadratic equation in standard form with solutions, $x = \frac{2}{3}$ and $x = -5$. Use integers for a , b and c .

51. Solve by factoring

$$x^2 - 8x = 0$$

$$x^2 - 10x = -25$$

$$x^2 = 72 - x$$

$$2x^2 + 5x + 3 = 0$$

$$5x^2 = 20x$$

$$x^2 = 81$$

Use synthetic substitution to find the given value for each function.

52. $2x^3 + 6x^2 - x + 15$ for $f(3)$

53. $3x^4 - 2x^3 + x^2 - 2$ for $f(-2)$

Given a polynomial, determine if the binomial is a factor.

54. $x^3 + 6x^2 - x - 30$, $x + 3$

55. $x^3 - 6x^2 + 11x - 6$, $x - 2$

Solve each system.

56.
$$\begin{aligned} 6x - y + 2z &= 8 \\ 2x + 3y - z &= -9 \\ 4x + 2y + 5z &= 1 \end{aligned}$$

57.
$$\begin{aligned} 5x - 6y + 2z &= 21 \\ 2x + 3y - 3z &= -9 \\ -3x + 9y - 4z &= -24 \end{aligned}$$

Pre-AP PreCalculus - Fall Semester Exam Review

Show all work on your own paper. Questions with *** are calculator questions

- Find the exact values of the six trig functions of an angle θ whose terminal side passes through the point $(-5, -7)$.

Find the exact value:

2. $\cos \frac{\pi}{4} \sin \frac{7\pi}{6} - \sin \frac{\pi}{6} \cos \frac{3\pi}{4}$

3. $\cot^2 \frac{11\pi}{6} - \csc^2 \frac{11\pi}{6}$

4. $\sin \frac{\pi}{2} + 6 \cos \frac{\pi}{3}$

5. $\frac{\cos \frac{5\pi}{3}}{\sin \frac{5\pi}{3}}$

6. $\sin^2 \frac{\pi}{6} + \cos^2 \frac{\pi}{6} + \tan^2 \frac{\pi}{6} - \sec^2 \frac{\pi}{6}$

- Convert 40° to radians

- Convert $\frac{\pi}{9}$ radians to degrees

- Find a positive and negative angle coterminal to 84°

- If $\sin \theta < 0$ and $\cot \theta < 0$, then θ must lie in which quadrant?

- ***Your cat is trapped on a tree branch 6.5 meters above the ground. Your ladder is only 6.7 meters long. If you place the ladder's tip on the branch, what angle will the ladder make with the ground?

- ***Commercial airliners fly at an altitude of about 10 kilometers. They start descending toward the airport when they are far away, so that they will not have to dive at a steep angle.

- If the pilot wants the plane's path to make an angle of 3° with the ground, at what horizontal distance from the airport must he start descending?

- If he starts descending a ground distance of 300 km from the airport, what angle will the plane's path make with the horizontal?

- ***While standing on a cliff 120 feet high, I see a sailboat at an angle of depression of 21° . What is the horizontal distance between the cliff and the sailboat?

Graph 1 cycle. Identify the period, amplitude, vertical shift and phase shift:

14. $y = -10 + 20 \sin 2\left(x - \frac{\pi}{8}\right)$

15. $y = -5 \cos \frac{1}{2}(x + \pi) + 3$

16. $y = 3 + 2 \cos \frac{1}{5}(x - \pi)$

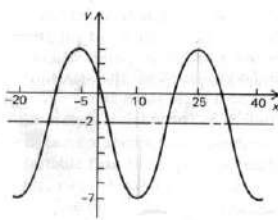
17. $y = 2 - 6 \sin \frac{\pi}{4}(x - 1)$

Write an equation for each graph as both sine and cosine:

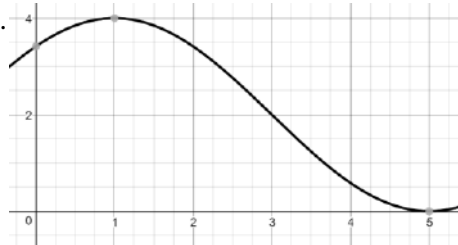
18.



19.



20.



21. ***Find each value correct to 3 decimal places.

$$f(x) = 5 + 2\cos\left(\frac{\pi}{4}(x-10)\right)$$

a.) $f(17.3)$

b.) Find the first three positive values for x if $f(x) = 6.7$

c.) What is the first positive x value that gives the maximum y value?

22. ***Researchers find a creature from an alien planet. Its body temperature is varying sinusoidally with time. 35 minutes after they start timing, it reaches a high of $120^\circ F$. 20 minutes after that it reaches its next low, $104^\circ F$.

a. Sketch a graph of this sinusoid.

b. Write an equation expressing temperature in terms of minutes since they started timing.

c. What was the temperature when they first started timing?

d. Find the first 3 times after they started timing at which the temperature was $114^\circ F$.

23. *** A spacecraft is in an elliptical orbit around the earth. At time $t=0$ hours, it is at its apogee (highest point) $d=1000$ km above the earth's surface. Fifty minutes later, it is at its perigee (closest point) $d=100$ km above the surface. Round to three decimal places.

a.) Write an equation for d in terms of t .

b.) Predict the first 3 positive values for t which the spacecraft is 200 km from the earth.

c.) In order to transmit back to earth, the spacecraft must be within 700 km of the surface. For how many consecutive minutes will the spacecraft be able to transmit?

24. Graph the parent function for each

a. $y = \tan x$

b. $y = \cot x$

c. $y = \csc x$

d. $y = \sec x$

Find values of x where $0 \leq x < 2\pi$

25. $\cos x = -\frac{\sqrt{2}}{2}$

27. $\tan^{-1}\left(\frac{\sqrt{3}}{3}\right) = x$

28. $\cos x = -1$

Find the exact value

29. $\sin^{-1}\left(-\frac{1}{2}\right)$

30. $\cos\left(\arcsin\left(-\frac{8}{17}\right)\right)$

31. $\cos^{-1}\left(\sin\frac{7\pi}{6}\right)$

Prove that each equation is an identity:

$$32. \frac{1}{1 + \cos x} = \csc^2 x - \csc x \cot x$$

$$33. \frac{\sin x}{1 - \cos x} + \frac{1 - \cos x}{\sin x} = 2 \csc x$$

$$34. \frac{1 - 3 \cos x - 4 \cos^2 x}{\sin^2 x} = \frac{1 - 4 \cos x}{1 - \cos x}$$

$$35. \tan x = \frac{1 - \cos 2x}{\sin 2x}$$

$$36. \text{ If } \sin A = \frac{1}{2}, \cos A > 0, \tan B = \frac{3}{4}, \text{ and } \sin B < 0, \text{ find } \sin(A + B).$$

$$37. \text{ If } \cos A = -\frac{6}{7} \text{ and } A \text{ is in Quadrant II, find } \sin 2A, \cos 2A, \text{ and } \tan 2A.$$

$$38. \text{ If } \cos \theta = -\frac{3}{5} \text{ and } 180^\circ < \theta < 270^\circ, \text{ find } \sin \frac{1}{2}\theta, \cos \frac{1}{2}\theta, \text{ and } \tan \frac{1}{2}\theta.$$

Solve each equation in its given domain:

$$39. 2 \sin \theta \cos \theta = \sqrt{2} \cos \theta \quad \{\text{real \# degrees}\}$$

$$40. 2 \cos^2 x - 5 \cos x + 2 = 0 \quad [0, 2\pi)$$

$$41. 4 \sin x \cos x = \sqrt{3} \quad [0, 2\pi)$$

$$42. \sin 2\theta \cos 64^\circ + \cos 2\theta \sin 64^\circ = \frac{\sqrt{3}}{2} \quad [0^\circ, 360^\circ)$$

$$43. \sin^2 x + 2 \sin x = 3 \quad [0, 2\pi)$$

$$44. \tan^2 x = 1 \quad [0, 2\pi)$$

This is a general overview of the trig you have learned this semester excluding unit 6 (oblique triangles of which you may use a calculator.) It would be a good idea to look back over all of your quizzes, homework, and test reviews.

Pre-Calculus: Course Study Guide

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❖ Prerequisites: Fundamental Concepts of Algebra

Algebraic Expressions and Real Numbers

- **Natural Numbers:** positive numbers
- **Whole Numbers:** number without fractions
- **Integers:** whole numbers and their opposite
- **Simple Fraction:** integer/natural
- **Absolute Value:** distance from 0
- **Evaluate** -> Expressions
 - $2x+3$; when $x=3$
- **Solve** -> Equations
 - $11=2x+3$
- **Intersection ($\cap$):** in both brackets at the same time
 - Ex: $A = \{1,3,4\}$ $B = \{3,4,5,6\}$ $A \cap B: \{3,4\}$
- **Union ($\cup$):** joining of things, join what's inside
- $\{ \}$: empty set; $\emptyset$: "null set" = no values

Exponent Rules For $a \neq 0, b \neq 0$	
Product Rule	$a^x \times a^y = a^{x+y}$
Quotient Rule	$a^x \div a^y = a^{x-y}$
Power Rule	$(a^x)^y = a^{xy}$
Power of a Product Rule	$(ab)^x = a^x b^x$
Power of a Fraction Rule	$\left(\frac{a}{b}\right)^x = \frac{a^x}{b^x}$
Zero Exponent	$a^0 = 1$
Negative Exponent	$a^{-x} = \frac{1}{a^x}$
Fractional Exponent	$a^{\frac{x}{y}} = \sqrt[y]{a^x}$

Radicals

- **Radical:** the main/primary root of what's underneath
 - $\sqrt{36}$ = square root
 - $\sqrt[3]{8}$ = cube root
 - $\sqrt{4x^2} = 2|x|$
 - the absolute value keeps the x from becoming negative
 - **IMPORTANT:** The only time you need the absolute value is when you take an **even root** of an **even power** and get an **odd result**. (EEO)
 - Examples

$$\blacksquare \sqrt{49x^2y^4} = 7|x^3|y^2$$

$$\blacksquare 2\sqrt[3]{5} + \sqrt[3]{5} = 3\sqrt[3]{5}$$

Factoring Formulas

- $(a+b)^2 = a^2 + 2ab + b^2$
 - $49x^2 + 126x + 81 = (7x + 9)^2$
- $(a+b)(a-b) = a^2 - b^2$ (difference of 2 squares)
- $a^2 + b^2$ = sum of 2 squares (can't factor) = prime
- $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$ = can never factor further than this!
- $a^3 + b^3$ = sum of 2 cubes
 - $(a+b)(a^2 - ab + b^2)$

Rational Expressions

- **Simplifying Rational Expressions**
 - 1) Factor the numerator and denominator completely
 - 2) Divide both the numerator and denominator by any common factors

$$\blacksquare \text{ Ex. } \frac{x^3+x^2}{x+1} = \frac{x^2(x+1)}{x+1} = \frac{x^2(x+1)}{(x+1)} \text{ -- cancel } (x+1)$$

$$\bullet = x^2, x \neq -1$$

- **Multiplying Rational Expressions**
 - 1) **Factor** all numerators and denominators completely
 - 2) **Divide** numerators and denominators by common factors
 - 3) **Multiply** the remaining factors in the numerators and multiply the remaining factors in the denominators

$$\frac{3x}{8y^2} \times \frac{y}{12} = \frac{\cancel{3}x}{8y^{\cancel{2}1}} \times \frac{\cancel{y}}{\cancel{12}_4} = \frac{x}{32y}$$

- **Dividing Rational Expressions**
 - 1) Find the answer by **inverting** the second divisor and multiplying using the steps above

$$\frac{7p}{p+2} \div \frac{p+2}{p} = \frac{7p}{p+2} \times \frac{p}{p+2} = \frac{7p^2}{(p+2)^2}$$

- **Adding/Subtracting Rational Expressions**

That Have Different Denominators

- 1) Find the **LCD** of the rational expression
- 2) Rewrite each rational expression as an equivalent expression whose denominator is the LCD.
- 3) Add/Subtract numerators by placing the resulting expression over the LCD.
- 4) If possible, simplify.

Finding Domain

• Ex. $\frac{(x-3)(2x+1)}{(x+1)(x-3)} = \frac{2x+1}{x+1}$ Domain = $(-\infty, -1) \cup (-1, \infty)$

Equations

- **Linear Equation:** an equation that is written in the form $ax+b=0$ where a and b are real numbers and $a \neq 0$

- **Solving Linear Equation**

- 1) Simplify both expressions
- 2) Group all variable terms on one side and constant terms on the other
- 3) Isolate the variable and solve

$$2(x + 2) - 5 = 3(x + 1)$$

$$2x - 1 = 3x + 3$$

$$-x - 1 = 3$$

$$-x = 4$$

$$x = -4$$

- **Solving Rational Equations**

- 1) Find the least common denominator
- 2) Use the distributive property and divide out common factors
- 3) Simplify

$$3x \cdot \left(\frac{5}{x} - \frac{1}{3} \right) = 3x \cdot \left(\frac{1}{x} \right)$$

$$3x \cdot \frac{5}{x} - 3x \cdot \frac{1}{3} = 3x \cdot \frac{1}{x}$$

$$15 - x = 3$$

$$-x = -12$$

$$x = 12$$

- **Solving Equation Involving Absolute Value**

- Isolate absolute value
- Split into 2 different equations
- Solve for both

$$|5 - 2x| - 11 = 0$$

$$|5 - 2x| = 11$$

$$5 - 2x = 11$$

$$-2x = 6$$

$$x = -3$$

$$5 - 2x = -11$$

$$-2x = -16$$

$$x = 8$$

- **Square Root Property**

- If $u^2 = d$, then $u = \sqrt{d}$ or $u = -\sqrt{d}$

- **Solving Radicals Containing nth Roots**

- 1) Arrange terms so one radical is isolated on one side of the equation.

- 2) Raise both sides of the equation to the nth power to eliminate the nth root.
- 3) Solve the resulting equation.
- Check all solutions in the original equation.

■ Ex. Solve: $\sqrt{2x-1} + 2 = x$

■ Step 1: $\sqrt{2x-1} + 2 = x \rightarrow \sqrt{2x-1} = x - 2$

■ Step 2: $(\sqrt{2x-1})^2 = (x-2)^2 \rightarrow 2x-1 = x^2 - 4x + 4$

■ Step 3: $2x - 1 = x^2 - 4x + 4$

■ $0 = x^2 - 6x + 5 \rightarrow 0 = (x-1)(x-5) \rightarrow x = 1 \text{ or } x = 5$

Linear Inequalities and Absolute Value Inequalities

- **Interval Notation:** represents subsets of real numbers
 - **Open Interval:** (a,b) represents the set of real numbers between, but not including a and b
 - **Closed Interval:** $[a,b]$ represents the set of real numbers between, and including a and b
 - **Infinite Interval:** (a, ∞) represents the set of numbers that are greater than a
 - **Infinite Interval:** $(-\infty, b]$ represents the set of real numbers that are less than or equal to b
- **Graphing Intersections and Unions**
 - 1) Graph each **interval** on a number line
 - 2A) To find the **intersection**, find the set of numbers on the number line where both graphs have the set in common
 - 2B) To find the **union**, take the portion of the number line representing the total collection of numbers in the two graphs
- **Solving an Absolute Value Inequality**

Properties of Unions of Sets	
Commutative Property	$A \cup B = B \cup A$
Associative Property	$(A \cup B) \cup C = A \cup (B \cup C)$
Identity Property	$A \cup \emptyset = \emptyset \cup A$
Distributive Property	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

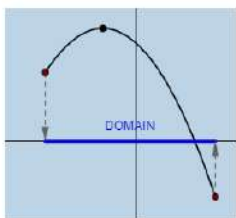
- If X is an algebraic expression and c is a positive number

- 1) The solutions of $|X| < c$ are the numbers that satisfy $-c < X < c$
- 2) The solutions of $|X| > c$ are the numbers that satisfy $X < -c$ or $X > c$

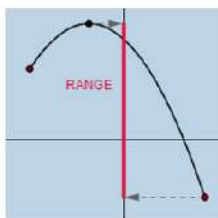
$$\begin{array}{lcl}
 |x - 4| < 7 & & \\
 \swarrow & & \searrow \\
 x - 4 < 7 & & x - 4 > -7 \\
 x - 4 + 4 < 7 + 4 & & x - 4 + 4 > -7 + 4 \\
 x < 11 & & x > -3 \\
 |9 - 4| = |5| = 5 & & |-2 - 4| = |-6| = 6 \\
 5 < 7 & & 6 < 7 \\
 & & \boxed{-3 < x < 11}
 \end{array}$$

❖ Graphs and Functions

Graphs and Graphing Utilities



Domain is all the possible x values of a function.



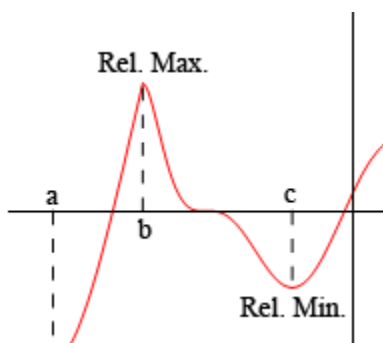
Range is all the possible y values of a function.

- **X- axis:** horizontal number line
- **Y- axis:** vertical number line
 - Each point corresponds to an ordered pair (x,y)

Basics of Functions and Their Graphs

- **Relation:** any set of ordered pairs
 - Set of first components is the **domain** (to find the domain, look for all inputs on the x -axis that correspond to the points on the graph). The set of second components is the **range** (to find the range, look for all the outputs on the y -axis that correspond to points on the graph)
- **Vertical Line Test:** if any vertical line intersects the graph in more than one point, the graph is not a function
- **Zeros of a function:** values of x for which $f(x) = 0$
 - A function can have more than one x -intercept, but at most one y -intercept

- **Difference Quotient:** $\frac{f(x+h)-f(x)}{h}, h \neq 0$
- **Piecewise Function:** defined by two (or more) equations over a specified domain



- **Relative Maximum:** the “peak” of the graph
- **Relative Minimum:** the “bottom” of the graph

Function	Even, Odd, or Neither?
$f(x) = 3x^2 + 8$	$f(-x) = 3(-x)^2 + 8 = 3x^2 + 8 = f(x)$ Even!
$f(x) = x^5 - 4x$	$f(-x) = (-x)^5 - 4(-x) = -x^5 + 4x$ $= -(x^5 - 4x) = -f(x)$ Odd!
$f(x) = 2x^2 - x - 1$	$f(-x) = 2(-x)^2 - (-x) - 1 = 2x^2 + x - 1$ $-f(x) = -(2x^2 - x - 1) = -2x^2 + x + 1$ $f(-x) \neq f(x) \neq -f(x)$ Neither!

- **Even Function:** $f(-x) = f(x)$; symmetric to the y-axis
- **Odd Function:** $f(-x) = -f(x)$; symmetric to the origin

Linear Functions and Slope

- To find slope: $\frac{y^2 - y^1}{x^2 - x^1}$
- **Point Slope:** $y - y_1 = m(x - x_1)$
- **General Form:** $Ax + By + C = 0$
- **Average Rate of Change:** $\frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$

Transformations of Functions

- **Vertical Shifts**

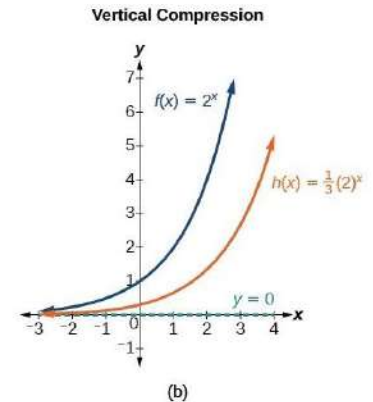
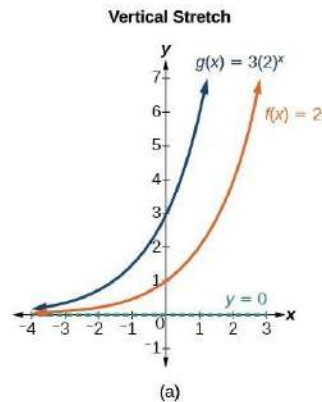
- The graph of $y = f(x) + c$ is the graph of $y = f(x)$ shifted c units **up**.
- The graph of $y = f(x) - c$ is the graph of $y = f(x)$ shifted c units **down**.

- **Horizontal Shifts**

- The graph of $y = f(x + c)$ is the graph of $y = f(x)$ shifted **left** c units.
- The graph of $y = f(x - c)$ is the graph of $y = f(x)$ shifted **right** c units.

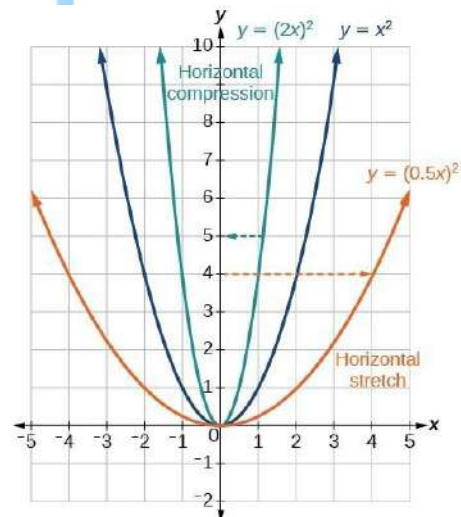
- **Vertically Stretching and Shrinking Graphs**

- In the graph $y = cf(x)$, if $c > 1$ the graph of $y = f(x)$ is **vertically stretched** by multiplying each of its y-coordinates by c .
- In the graph $y = cf(x)$, if $0 < c < 1$, the graph $y = f(x)$ is **vertically shrunk** by multiplying each of its y-coordinates by c .



- **Horizontally Stretching and Shrinking**

- In the graph $y = f(cx)$, if $c > 1$ the graph of $y = f(x)$ is **horizontally shrunk** by dividing each of its x-coordinates by c .
- In the graph of $y = f(cx)$, if $0 < c < 1$, the graph of $y = f(x)$ is **horizontally stretched** by dividing each of its x-coordinates by c .

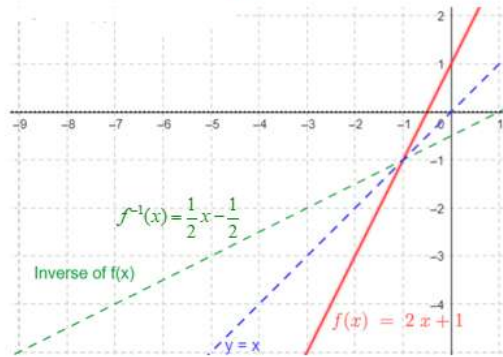


Inverse Functions

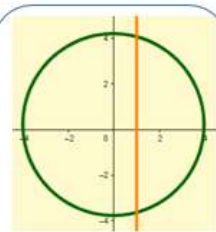
- **$f(g(x)) = x$ and $g(f(x)) = x$** , the function g is the inverse of function f which means f and g are inverses of each other

Find the inverse of the function $f(x) = 2x + 1$

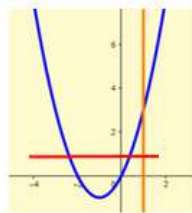
$$\begin{aligned} f(x) &= 2x + 1 \\ y &= 2x + 1 \\ y - 1 &= 2x \\ \frac{y - 1}{2} &= x \\ y &= \frac{x - 1}{2} \\ y &= \frac{1}{2}x - \frac{1}{2} \\ f^{-1}(x) &= \frac{1}{2}x - \frac{1}{2} \end{aligned}$$



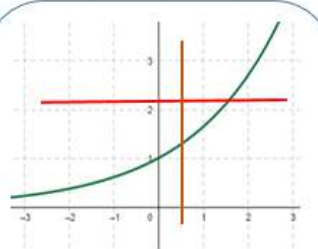
$f(x)$ and $f^{-1}(x)$ are mirror images about the line $y = x$



Fail Vertical Line Test
Not a Function



Pass Vertical Line Test
Fail Horizontal Line Test
Not a One-to-One Function



Pass Vertical Line Test
Pass Horizontal Line Test
A One-to-One Function
An invertible function that has an inverse

- **Finding a Function's Inverse**
 - Switch x and y , then solve for y
- **Horizontal Line Test:** If function f has an inverse that is a function f^{-1} , if there is no horizontal line that intersects the graph of the function f at more than one point.

Distance and Midpoint Formulas: Circles

- **Distance Formula:** $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- **Midpoint Formula:** $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$

- **Standard Form of Circle:** $(x - h)^2 + (y - k)^2 = r^2$
- **General Form of Circle Equation:** $x^2 + y^2 + Dx + Ey + F = 0$

❖ Polynomial and Rational Functions

Complex Numbers

- **Imaginary Unit i** is defined as $i =$

$$\sqrt{-1}, \text{ where } i^2 = -1$$

- In $a + bi$, a is the real part and b is the imaginary part
- **Complex Numbers:** real numbers a and b , and i , the imaginary unit
- **Pure Imaginary Number:** imaginary number in the form of bi
- **Adding and Subtracting Complex Numbers**

- $(a + bi) + (c + di) = (a + c) + (b + d)i$

- Add complex numbers by adding the real parts, adding the imaginary parts, and expressing the sum as a complex number

- $(a + bi) - (c + di) = (a - c) + (b - d)i$

- Subtract complex numbers by subtracting the real parts, subtracting the imaginary parts, and expressing the sum as a complex number

- **Multiplying and Dividing Complex Numbers**

- For multiplication, FOIL, first, outer, inner, last, then solve.

$$\begin{aligned}(a + bi)(c + di) &= ac + adi + bc i + bd i^2 \\ &= ac + (ad + bc)i - bd \\ &= \underbrace{ac - bd}_{\text{Real}} + \underbrace{(ad + bc)i}_{\text{Imagine}}\end{aligned}$$

- For division, multiply the numerator and the denominator by the complex conjugate of the denominator.

$$\begin{aligned}\frac{(a + bi)}{(c + di)} \cdot \frac{(c - di)}{(c - di)} &= \frac{ac - adi + bc i - bd i^2}{c^2 - d^2 i^2} \\ &= \frac{ac + bd + (-ad + bc)i}{c^2 + d^2}\end{aligned}$$

Use the FOIL method.

Combine any imaginary terms, then combine real terms.

Example 1: Add $4 + 2i$ and $6 + 3i$

$$\begin{aligned}4 + 2i + 6 + 3i \\ &= (4 + 6) + (2i + 3i) \\ &= 10 + 5i\end{aligned}$$

Example 2: Subtract $(4 + 2i) - (6 + 3i)$

$$\begin{aligned}4 + 2i - 6 - 3i \\ &= (4 - 6) + 2i - 3i \\ &= -2 - i\end{aligned}$$

- **Complex Conjugate** of the number $a + bi$ is $a - bi$
- **Principle Square Root:** for any positive real number b , the principal square root of the negative number $-b$ is defined by $\sqrt{-b} = i\sqrt{b}$

Quadratic Functions

- **Quadratic Function** is of the form: $f(x) = ax^2 + bx + c, a \neq 0$
- **Standard Form** of Quadratic Function: $f(x) = a(x - h)^2 + k, a \neq 0$
 - **Vertex:** (h, k)
- **Maximum and Minimum of Quadratic Functions**
 - Function $f(x) = ax^2 + bx + c$
 - If $a > 0$, then f has a minimum that occurs at $x = -\frac{b}{2a}$, minimum value $f(-\frac{b}{2a})$
 - If $a < 0$, then f has the maximum that occurs at $x = -\frac{b}{2a}$, maximum value $f(-\frac{b}{2a})$

Polynomial Functions and Their Graphs

- **Polynomial Function:** a function comprising more than one power function where the coefficients are assumed to not equal zero. The term with the highest degree is the leading term.

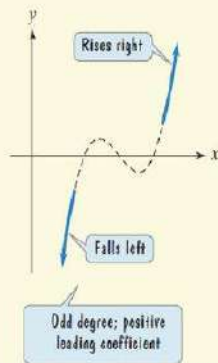
The Leading Coefficient Test for

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x + a_0 \quad (\text{continued})$$

1. For n odd:

If the leading coefficient is positive, the graph falls to the left and rises to the right. ($\swarrow, \nearrow$)

$$a_n > 0$$



If the leading coefficient is negative, the graph rises to the left and falls to the right. ($\nwarrow, \searrow$)

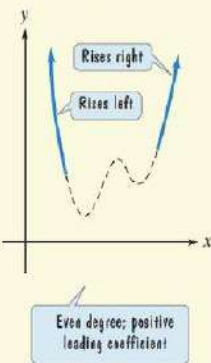
$$a_n < 0$$



2. For n even:

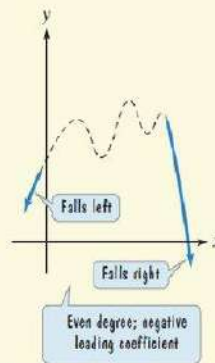
If the leading coefficient is positive, the graph rises to the left and rises to the right. ($\nwarrow, \nearrow$)

$$a_n > 0$$



If the leading coefficient is negative, the graph falls to the left and falls to the right. ($\swarrow, \searrow$)

$$a_n < 0$$



- **Even multiplicity:** the graph touches the x-axis and turns around at r
- **Odd multiplicity:** the graph crosses the x-axis at r
- **Graphing a Polynomial**

Function

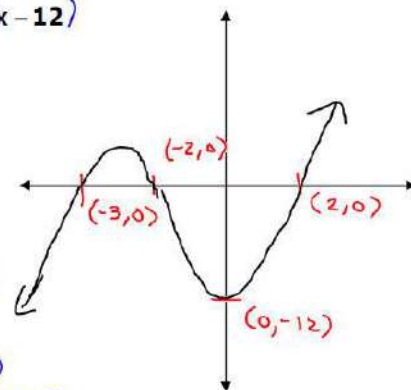
- 1) Use the **Leading Coefficient Test** to determine graph's end behavior
- 2) Find **x-intercept** by setting $f(x) = 0$ and solving the resulting polynomial equation.
- 3) Find the **y-intercept** by computing $f(0)$
- 4) Use **symmetry**, if applicable to help draw the graph.
 - Y-axis symmetry: $f(-x) = f(x)$
 - Origin symmetry: $f(-x) = -f(x)$
- 5) Use the fact that the maximum number of turning points of the graph is $n-1$ to

$$P(x) = x^3 + 3x^2 - 4x - 12$$

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Section 4.1

End Behavior $\nearrow$
+ odd
y-int $x=0$ $(0, -12)$

$$\begin{aligned} x^2(x+3) - 4(x+3) &= 0 \\ (x+3)(x^2-4) &= 0 \\ x+3 &= 0 \quad (x+2)(x-2) = 0 \\ x &= -3, \quad x = -2, \quad x = 2 \end{aligned}$$



check whether its drawn correctly

- **Intermediate Value Theorem:** If f is a polynomial function and $f(a)$ and $f(b)$ have opposite signs, there is at least one value of c between a and b for which $f(c) = 0$.

Dividing Polynomials

- **Division Algorithm**

- $f(x) = d(x) * q(x) + r(x)$
- $f(x)$ = dividend
- $d(x)$ = divisor
- $q(x)$ = quotient
- $r(x)$ = remainder

$$\begin{array}{r}
 x + 2 \overline{) 2x^3 - 3x^2 + 4x + 5} \\
 \underline{2x^2} \\
 x + 2 \overline{) 2x^3 - 3x^2 + 4x + 5} \\
 \underline{2x^2} \\
 x + 2 \overline{) 2x^3 - 3x^2 + 4x + 5} \\
 \underline{-(2x^3 + 4x^2)} \\
 -7x^2 + 4x \\
 \underline{2x^2 - 7x} \\
 x + 2 \overline{) 2x^3 - 3x^2 + 4x + 5} \\
 \underline{-(2x^3 + 4x^2)} \\
 -7x^2 + 4x \\
 \underline{-(-7x^2 + 14x)} \\
 18x + 5 \\
 \underline{2x^2 - 7x + 18} \\
 x + 2 \overline{) 2x^3 - 3x^2 + 4x + 5} \\
 \underline{-(2x^3 + 4x^2)} \\
 -7x^2 + 4x \\
 \underline{-(-7x^2 + 14x)} \\
 18x + 5 \\
 \underline{-18x + 36} \\
 -31
 \end{array}$$

Set up the division problem.

$2x^3$ divided by x is $2x^2$.

Multiply $x + 2$ by $2x^2$.

Subtract.

Bring down the next term.
 $-7x^2$ divided by x is $-7x$.

Multiply $x + 2$ by $-7x$.
 Subtract. Bring down the next term.

$18x$ divided by x is 18 .

Multiply $x + 2$ by 18 .
 Subtract.

- **Long Division of Polynomials**

- 1) Set up long division
- 2) Divide the 1st term of the dividend with the divisor
- 3) Multiply by the divisor
- 4) Write the answer and subtract
- 5) Bring down the next number to the right
- 6) Repeat Step 2
- 7) Write final answer

- **Remainder Theorem**

- If the polynomial $f(x)$ is divided by $x - c$, then the remainder is $f(c)$.

- **Factor Theorem**

- Let $f(x)$ be a polynomial
 - a) If $f(c) = 0$, then $x - c$ is a factor of $f(x)$
 - b) If $x - c$ is a factor of $f(x)$, then $f(c) = 0$

Zeros of Polynomial Functions

- **Rational Zero Theorem**

Rational Root Theorem

The **rational roots theorem** tells you a list of possible rational roots for a given a polynomial function.

$$\text{Possible Rational Roots} = \frac{\text{factors of the constant}}{\text{factors of the lead coefficient}}$$

Example:

What are the possible rational roots of $6x^3 + 8x^2 - 7x - 3$

The leading coefficient is 6.

The constant term is -3.

The factors of 6 are $\pm 1, \pm 2, \pm 3, \pm 6$. The factors of -3 are $\pm 1, \pm 3$.

$$\begin{aligned} \text{Possible Rational Roots} &= \frac{\pm 1, \pm 3}{\pm 1, \pm 2, \pm 3, \pm 6} \\ &= \pm 1, \pm \frac{1}{2}, \pm \frac{1}{3}, \pm \frac{1}{6}, \pm 3, \pm \frac{3}{2} \end{aligned}$$

• Number of Roots

- If $f(x)$ is a polynomial degree $n \geq 1$, counting multiple roots separately, the equation $f(x) = 0$ has n roots.

• Descartes's Rule of Signs

- The number of positive real zeros of f equals the number of sign changes of $f(x)$ or is less than that number by an even integer. The number of negative real zeros of f applies a similar statement to $f(-x)$.

Determine the number of possible positive and negative real zeros.

$$g(x) = 2x^6 - 5x^4 - 3x^3 + 7x^2 + 2x + 5$$

Solution:

$g(x)$ has real coefficients and the constant term is nonzero.

$$g(x) = 2x^6 - 5x^4 - 3x^3 + 7x^2 + 2x + 5 \quad \text{2 sign changes in } g(x)$$

The number of possible positive real zeros is either 2 or 0.

$$\begin{aligned} g(-x) &= 2(-x)^6 - 5(-x)^4 - 3(-x)^3 + 7(-x)^2 + 2(-x) + 5 \\ &= 2x^6 - 5x^4 + 3x^3 + 7x^2 - 2x + 5 \quad \text{4 sign changes in } g(-x) \end{aligned}$$

The number of possible negative real zeros is either 4, 2, or 0.

Number of possible positive real zeros	2	2	2	0	0	0
Number of possible negative real zeros	4	2	0	4	2	0
Number of nonreal zeros	0	2	4	2	4	6
Total (including multiplicities)	6	6	6	6	6	6

Rational Functions and Their Graphs

• Arrow Notation

- $x \rightarrow a^+$: x approaches a from the right

- $x \rightarrow a^-$: x approaches a from the left
- $x \rightarrow \infty$: x approaches infinity
- $x \rightarrow -\infty$: x approaches negative infinity
- **Vertical Asymptote**
 - If $f(x)$ increases or decreases without bound as x approaches a
- **Horizontal Asymptote**
 - If $f(x)$ approaches b as x increases or decreases without bound
- **Graphing Rational Functions** $f(x) = \frac{p(x)}{q(x)}$
 - Determine whether the graph has **symmetry**
 - $f(-x) = f(x)$: y-axis symmetry
 - $f(-x) = -f(x)$: origin symmetry
 - Find the **y-intercept** by evaluating $f(0)$
 - Find the **x-intercept** by solving the equation $p(x)=0$
 - Find any **vertical asymptotes** by solving the equation $q(x) = 0$
 - Find the **horizontal asymptote** by using the rule for determining the horizontal asymptote of a rational function.
 - Plot at least one point between and beyond each x-intercept and vertical asymptote.
 - Use info above to graph the function between the asymptotes

❖ Exponential and Logarithmic Functions

Exponential Functions

- **Parent Function:** $f(x) = b^x$, where b is base and $b > 0, b \neq 1$
- **Natural exponential function:** $f(x) = e^x$

- **Irrational number (natural base):** $e \approx 2.7183$, e is the value that $(1 + \frac{1}{n})^n$

Logarithmic Functions

- **Logarithmic Functions:** for $x > 0$ and $b > 0, b \neq 1$

Logarithmic Properties	
Product Rule	$\log_a(xy) = \log_a x + \log_a y$
Quotient Rule	$\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$
Power Rule	$\log_a x^p = p \log_a x$
Change of Base Rule	$\log_a x = \frac{\log_b x}{\log_b a}$
Equality Rule	If $\log_a x = \log_a y$ then $x = y$

Logarithmic Function

$\log_a x = y$ means $a^y = x$

exponent
base

$a > 0, a \neq 1, y \neq 0$

Example:
 $\log_2 8 = 3$ means $2^3 = 8$

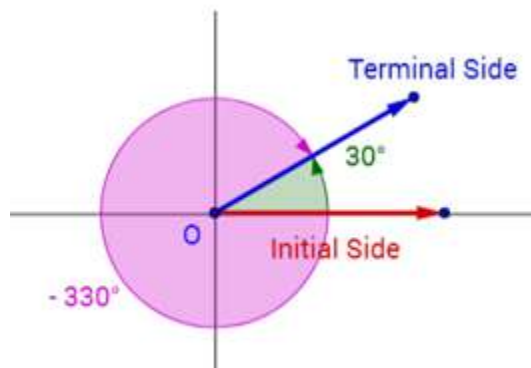
Exponential and Logarithmic Equations

- **Exponential equation:** equation containing a variable in an exponent
- **Solving Exponential Equations by Expressing Each Side as a Power of the Same Base**
 - If $b^M = b^N$, then $M = N$
- **Using Natural Logarithms to Solve Exponential Equations**
 - 1) Isolate the exponential expression
 - 2) Take the natural logarithm on both sides of the equation
 - 3) Simplify using one of the following
 - $\ln(b^x) = x \ln(b)$ or $\ln e^x = x$
 - 4) Solve for the variable
- **Using the Definition of a Logarithm to Solve Logarithmic Equations**
 - 1) Express the equation in the form $\log_b M = c$.
 - 2) Use the definition to rewrite the equation in exponential form $\log_b M = c$ means $b^c = M$
 - 3) Solve for the variable.

❖ Trigonometric Functions

Angles and Radian Measure

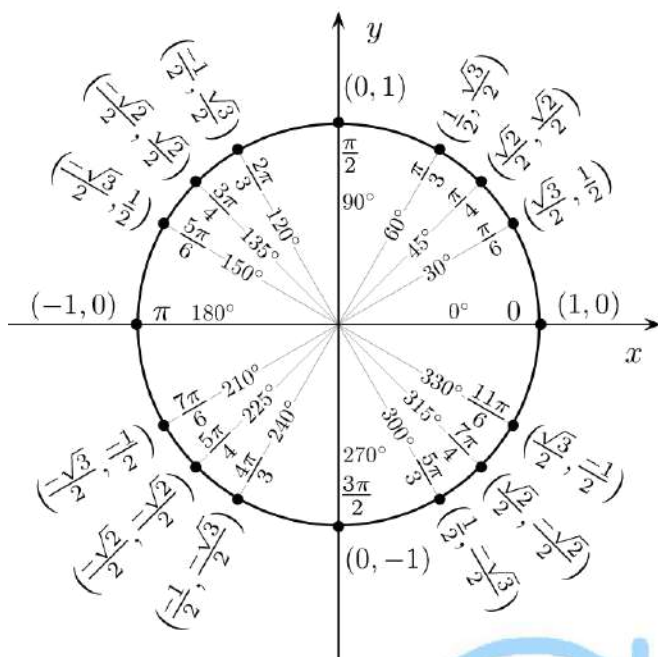
- **Angle:** two rays with a common endpoint called the vertex
- **Quadrantal Angle:** angle with its terminal side on the x-axis or the y-axis
- **Radian Measure:** $\theta = \frac{s}{r} \text{ radians}$
- To **convert from degrees to radians**, multiply by $\frac{\pi \text{ radians}}{180^\circ}$
- To **convert from radians to degrees**, multiply by $\frac{180^\circ}{\pi \text{ radians}}$
- **Coterminal Angles:** angles with the same initial and terminal sides
- **Length of a Circular Arc:** $s = r\theta$



The Unit Circle

- **Definitions of the Trigonometric Functions in Terms of a Unit Circle**
 - If t is a real number and $P = (x, y)$ is a point on the unit circle that corresponds to t , then

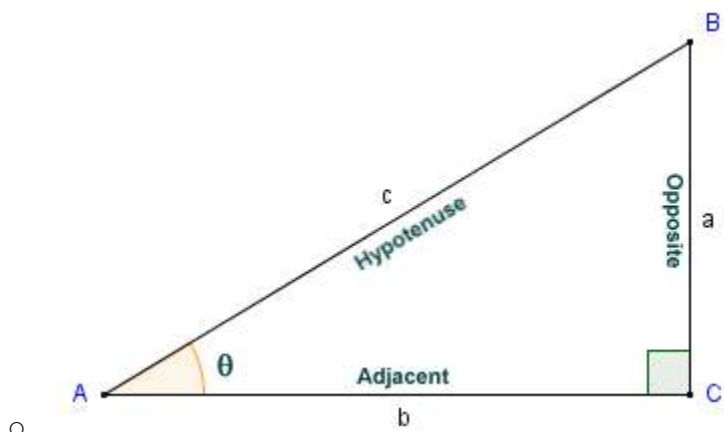
$\sin t = y$	$\cos t = x$	$\tan t = \frac{y}{x}, x \neq 0$
$\csc t = \frac{1}{y}, y \neq 0$	$\sec t = \frac{1}{x}, x \neq 0$	$\cot t = \frac{x}{y}, y \neq 0$



- **Domain and Range of Sine and Cosine Functions**
 - **Domain** to sine and cosine function $(-\infty, \infty)$, set of all real numbers
 - **Range** $[-1, 1]$
- **Even and Odd Trigonometric Functions**
 - **Even**
 - $\cos(-t) = \cos t$ and $\sec(-t) = \sec t$
 - **Odd**
 - $\sin(-t) = -\sin t, \tan(-t) = -\tan(t), \csc(-t) = -\csc t, \cot(-t) = -\cot(t)$
- **Quotient Identities**
 - $\tan t = \frac{\sin t}{\cos t}$ and $\cot t = \frac{\cos t}{\sin t}$
- **Pythagorean Identities**
 - $\sin^2 t + \cos^2 t = 1$ $1 + \tan^2 t = \sec^2 t$ $1 + \cot^2 t = \csc^2 t$
- **Periodic Functions:** a function that repeats its values at regular intervals
 - **Periodic Properties for Sine and Cosine (period 2π)**
 - $\sin(t + 2\pi) = \sin t$ and $\cos(t + 2\pi) = \cos t$
 - **Periodic Properties for Tangent and Cotangent (period π)**
 - $\tan(t + \pi) = \tan t$ and $\cot(t + \pi) = \cot t$

Right Triangle Trig

- Right Triangle Definitions for Trig Functions**



$\sin \theta = \frac{a}{c}$	$\cos \theta = \frac{b}{c}$	$\tan \theta = \frac{a}{b}$
$\csc \theta = \frac{c}{a}$	$\sec \theta = \frac{c}{b}$	$\cot \theta = \frac{b}{a}$

- Cofunction Identities**

$\sin \theta = \cos (90^\circ - \theta)$	$\cos \theta = \sin (90^\circ - \theta)$	$\tan \theta = \cot (90^\circ - \theta)$
$\cot \theta = \tan (90^\circ - \theta)$	$\sec \theta = \csc (90^\circ - \theta)$	$\csc \theta = \sec (90^\circ - \theta)$

Trigonometric Functions of Any Angle

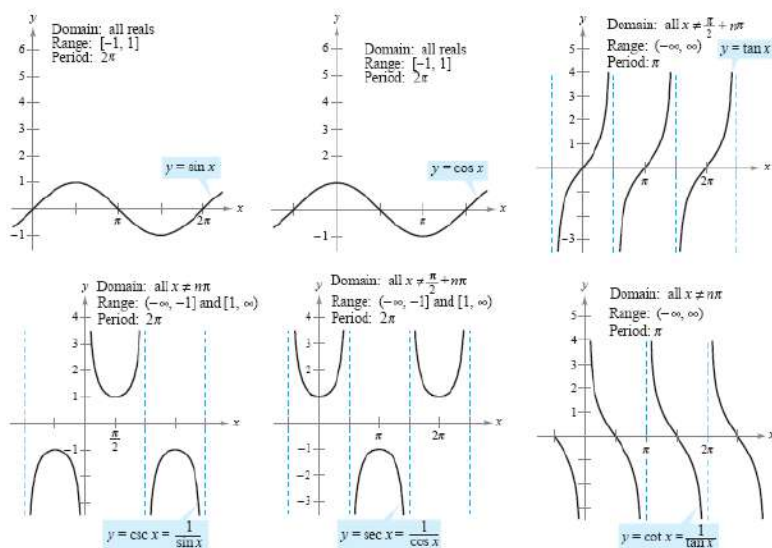
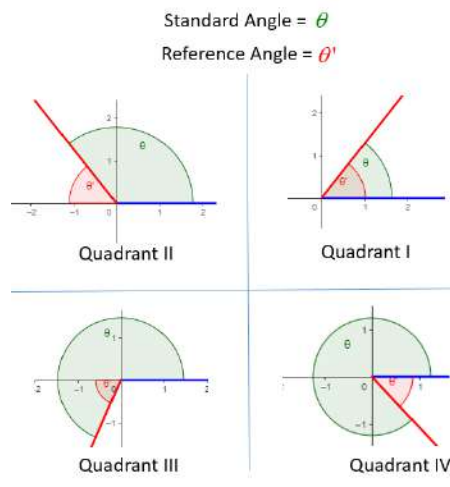
- If $r = \sqrt{x^2 + y^2}$ is the distance from (0,0) to (x,y), the six trig functions of θ are

$\sin \theta = \frac{y}{r}$	$\csc \theta = \frac{r}{y}, y \neq 0$
$\cos \theta = \frac{x}{r}$	$\sec \theta = \frac{r}{x}, x \neq 0$
$\tan \theta = \frac{y}{x}, x \neq 0$	$\cot \theta = \frac{x}{y}, y \neq 0$

- **Reference Angle:** positive acute angle between the terminal side and x-axis
- **Find Reference Angles for Angles Greater Than 360° (2π) or Less Than -360° (-2π)**

- Find a positive angle α less than 360° that is coterminal with the given angle.
- Draw α in standard position.
- Use the drawing to find the reference angle for the given angle.

Graphs of Trig Functions



The graphs of the six trigonometric functions

- The graph of $y = A \sin(Bx + C)$ can be obtained using **amplitude** $|A|$, **period** $\frac{2\pi}{b}$, and **phase shift** $\frac{C}{B}$.
- The graph of $y = A \cos(Bx - C)$ can be obtained using **amplitude** $|A|$, **period** $\frac{2\pi}{b}$, and **phase shift** $\frac{C}{B}$.
- $y = A \sin(Bx + C) + D$ and $y = A \cos(Bx - C) + D$, the constant D can cause **vertical shifts**. If $D > 0$, you shift **upward**. If $D < 0$, you shift **downward**.

Inverse Trig Functions

- **Inverse Sine Function:** inverse of the restricted sine function $y = \sin x, -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ which means $y = \sin^{-1}x$ means $\sin y = x$

- **Finding Exact Values of $\sin^{-1}x$**

- 1) Let $\theta = \sin^{-1}x$
- 2) Rewrite $\theta = \sin^{-1}x$ as $\sin\theta = x$, where $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$
- 3) Use the table to find the exact values that satisfies $\sin\theta = x$

θ	$-\frac{\pi}{2}$	$-\frac{\pi}{3}$	$-\frac{\pi}{4}$	$-\frac{\pi}{6}$	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin \theta$	-1	$-\frac{\sqrt{3}}{2}$	$-\frac{\sqrt{2}}{2}$	$-\frac{1}{2}$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1

- **Inverse Cosine Function:** inverse of the restricted cosine function $y = \cos x, 0 \leq x \leq \pi$ which means $y = \cos^{-1}x$ means $\cos y = x$

- **Finding Exact Values of $\cos^{-1}x$**

- 1) Let $\theta = \cos^{-1}x$
- 2) Rewrite $\theta = \cos^{-1}x$ as $\cos\theta = x$, where $0 \leq \theta \leq \pi$
- 3) Use the table to find the exact values

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{3}}{2}$	-1

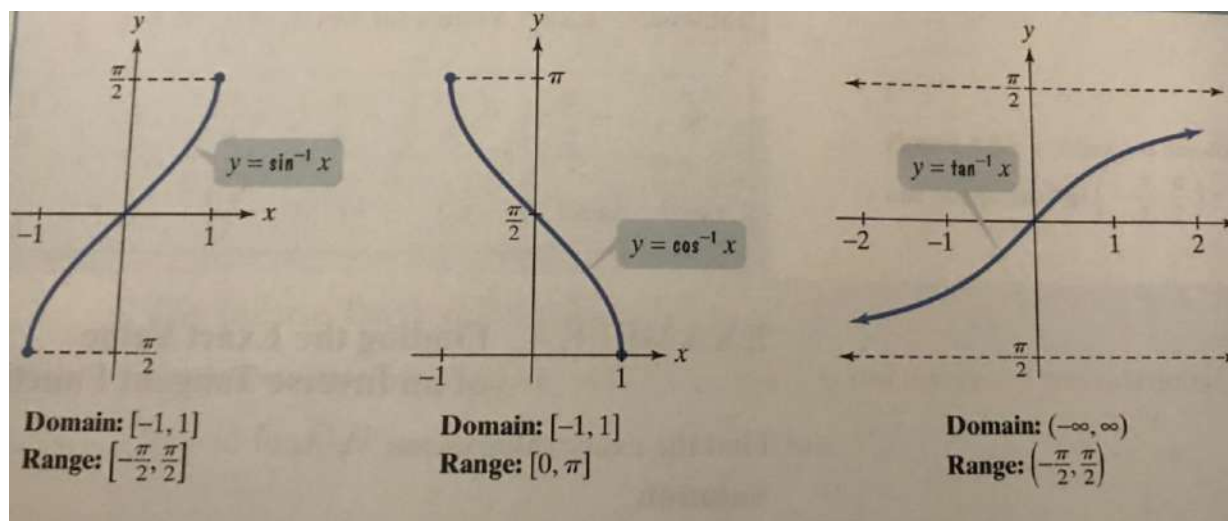
- **Inverse Tangent Function:** inverse of the restricted tangent function $y = \tan x, -\frac{\pi}{2} < x < \frac{\pi}{2}$ which means $y = \tan^{-1}x$ means $\tan y = x$

- **Finding Exact Values of $\tan^{-1}x$**

- 1) Let $\theta = \tan^{-1}x$
- 2) Rewrite $\theta = \tan^{-1}x$ as $\tan \theta = x$, where $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$
- 3) Use the table to find exact values

θ	$-\frac{\pi}{2}$	$-\frac{\pi}{3}$	$-\frac{\pi}{4}$	$-\frac{\pi}{6}$	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\tan \theta$	undef.	$-\sqrt{3}$	-1	$-\frac{\sqrt{3}}{3}$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undef.

• Graphs of Three Basic Inverse Trig Functions



❖ Analytic Trigonometry

Verifying Trig Identities

• Fundamental Trig Identities

○ Reciprocal Identities

$\sin x = \frac{1}{\csc x}$	$\cos x = \frac{1}{\sec x}$	$\tan x = \frac{1}{\cot x}$
$\csc x = \frac{1}{\sin x}$	$\sec x = \frac{1}{\cos x}$	$\cot x = \frac{1}{\tan x}$

○ Quotient Identities

$\tan x = \frac{\sin x}{\cos x}$	$\cot x = \frac{\cos x}{\sin x}$
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○ Pythagorean Identities

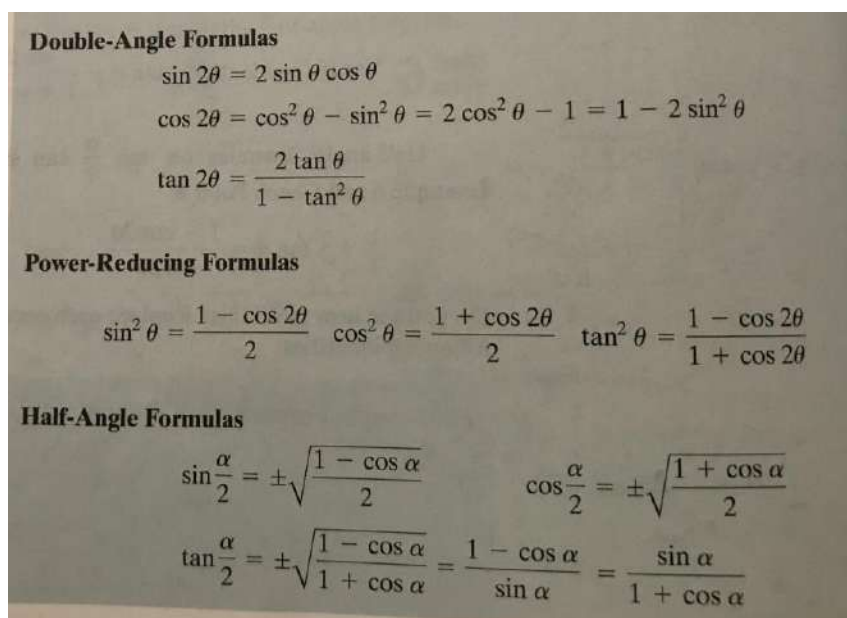
$\sin^2 x + \cos^2 x = 1$	$1 + \tan^2 x = \sec^2 x$	$1 + \cot^2 x = \csc^2 x$
---------------------------	---------------------------	---------------------------

○ Even-Odd Functions

$\sin(-x) = -\sin x$	$\cos(-x) = \cos x$	$\tan(-x) = -\tan x$
$\csc(-x) = -\csc x$	$\sec(-x) = \sec x$	$\cot(-x) = -\cot x$

• **Principal Trig Identities**

$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$	$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$
$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$	$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$
$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$	$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$



○ Product To Sum Formula

$\sin \alpha \sin \beta = \frac{1}{2} [\cos(\alpha - \beta) - \cos(\alpha + \beta)]$
$\cos \alpha \cos \beta = \frac{1}{2} [\cos(\alpha - \beta) + \cos(\alpha + \beta)]$
$\sin \alpha \cos \beta = \frac{1}{2} [\sin(\alpha + \beta) - \sin(\alpha - \beta)]$
$\cos \alpha \sin \beta = \frac{1}{2} [\sin(\alpha + \beta) - \sin(\alpha - \beta)]$

$$\begin{aligned}\sin A + \sin B &= 2 \sin \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right) \\ \sin A - \sin B &= 2 \cos \left(\frac{A+B}{2} \right) \sin \left(\frac{A-B}{2} \right) \\ \cos A + \cos B &= 2 \cos \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right) \\ \cos A - \cos B &= -2 \sin \left(\frac{A+B}{2} \right) \sin \left(\frac{A-B}{2} \right)\end{aligned}$$

→ Sum to Product Formulas

❖ Conic Sections

- **The Ellipse:** is the set of all points P in a plane the sum of whose distances from two fixed points is constant
 - **Foci:** the two fixed points
 - **Center:** midpoint of the segment connecting the foci
- **Standard Form Of Ellipse**
 - $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ or $\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$
- **Standard Form of Equations for Ellipses**

Equation	Center	Major Axis	Vertices	Graph
$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$ Endpoints of major axis are a units right and a units left of center. Foci are c units right and c units left of center, where $c^2 = a^2 - b^2$. $a^2 > b^2$	(h, k)	Parallel to the x-axis, horizontal	$(h-a, k)$ $(h+a, k)$	
$\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1$ Endpoints of the major axis are a units above and a units below the center. Foci are c units above and c units below the center, where $c^2 = a^2 - b^2$. $a^2 > b^2$	(h, k)	Parallel to the y-axis, vertical	$(h, k-a)$ $(h, k+a)$	

-
- **The Hyperbola:** a set of points in a plane the difference of whose distances from two fixed points, called foci is constant

- **Vertices:** line through the foci that intersects at two points
- **Standard Forms of the Equations of a Hyperbola**

■ $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ or $\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$

- **Standard Forms of Hyperbolas Centered at (h,k)**

Equation	Center	Transverse Axis	Vertices	Graph
$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$ Vertices are a units right and a units left of center. Foci are c units right and c units left of center, where $c^2 = a^2 + b^2$.	(h, k)	Parallel to the x-axis, horizontal	$(h - a, k)$ $(h + a, k)$	
$\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$ Vertices are a units above and a units below the center. Foci are c units above and c units below the center, where $c^2 = a^2 + b^2$.	(h, k)	Parallel to the y-axis, vertical	$(h, k - a)$ $(h, k + a)$	

- **The Parabola:** set of all points in a plane that are equidistant from a fixed line.
 - **Directrix:** a fixed point
 - **Focus:** not on the line
 - **Latus Rectum:** line segment that passes through its focus, parallel to the directrix, the endpoints are located on the parabola

■ Latus rectum is $|4p|$

- **Standard Form of the Equation of a Parabola**

○ $y^2 = 4px$ or $x^2 = 4py$

Equation	Vertex	Axis of Symmetry	Focus	Directrix	Description
$(y - k)^2 = 4p(x - h)$	(h, k)	Horizontal	$(h + p, k)$	$x = h - p$	If $p > 0$, opens to the right. If $p < 0$, opens to the left.
$(x - h)^2 = 4p(y - k)$	(h, k)	Vertical	$(h, k + p)$	$y = k - p$	If $p > 0$, opens upward. If $p < 0$, opens downward.

❖ Matrices and Determinants

Method of Gaussian Elimination 3x3 Example

$$\begin{aligned} 4x - 3y + z &= -8 \\ -2x + y - 3z &= -4 \\ x - y + 2z &= 3 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 4 & -3 & 1 & -8 \\ -2 & 1 & -3 & -4 \\ 1 & -1 & 2 & 3 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 3 \\ 0 & -1 & 1 & 2 \\ 0 & 1 & -7 & -20 \end{array} \right]$$

Solve for x , y , and z

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

$$\begin{aligned} 1x + 0y + 0z &= -2 \\ 0x + 1y + 0z &= 1 \\ 0x + 0y + 1z &= 3 \end{aligned}$$

$$x = 2 \quad y = 1 \quad z = 3$$

- **Augmented Matrices:** has a vertical bar separating the columns of the matrix into 2 groups
- **Row-Echelon Form:** matrix with 1s down the main diagonal and 0s below the 1s
- **Gaussian Elimination:** process used to solve linear systems using matrix row operations
- **Gauss-Jordan Elimination:** reduced row-echelon form, the process for a matrix with 1s down the main diagonal and 0s in every position above and below each 1 is found

Example: The system of equations $\begin{cases} x + y + z = 3 \\ 2x + 3y + 7z = 0 \\ x + 3y - 2z = 17 \end{cases}$ has augmented matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 2 & 3 & 7 & 0 \\ 1 & 3 & -2 & 17 \end{array} \right]$$

Row operations can be used to express the matrix in reduced row-echelon form.

$$\begin{aligned} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 2 & 3 & 7 & 0 \\ 1 & 3 & -2 & 17 \end{array} \right] &\rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & 5 & -6 \\ 0 & 2 & -3 & 14 \end{array} \right] \\ &\rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -4 & 9 \\ 0 & 1 & 5 & -6 \\ 0 & 0 & -13 & 26 \end{array} \right] \\ &\rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -4 & 9 \\ 0 & 1 & 5 & -6 \\ 0 & 0 & 1 & -2 \end{array} \right] \\ &\rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & -2 \end{array} \right] \end{aligned}$$

- **Matrix Addition and Subtraction:** matrices of the same order are added or subtracted by adding or subtracting.

- **Properties of Matrix Addition**

- 1) $A+B = B+A$
- 2) $(A+B) + C = A + (B+C)$
- 3) $A+0 = 0+A = A$
- 4) $A + (-A) = (-A) + A = 0$

- **Scalar Multiplication:** product of a real number and a matrix

$$2 \cdot \begin{bmatrix} 10 & 6 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 2 \cdot 10 & 2 \cdot 6 \\ 2 \cdot 4 & 2 \cdot 3 \end{bmatrix}$$

- **Properties of Scalar Multiplication**

- $(cd)A = c(dA)$
- $1A=A$
- $c(A+B) = cA+cB$
- $(c+d)A = cA+dA$

- **Properties of Matrix Multiplication**

- $(AB)C= A(BC)$
- $A(B+C)= AB+ AC$
- $(A+B) C = AC + BC$
- $c(AB) = (cA) B$

- **Finding Multiplicative Inverses for Invertible Matrices**

Use a graphing utility with matrix capabilities, or

a. If the matrix is 2×2 : The inverse of $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

- **Finding the Determinant of a 2x2 Matrix**

The determinant of a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is $ad - bc = 0$

The determinant of a 3×3 matrix $\begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix}$ is

$$a_1 \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & c_1 \\ b_3 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix} = 0$$

$$a_1(b_2c_3 - b_3c_2) - a_2(b_1c_3 - b_3c_1) + a_3(b_1c_2 - b_2c_1) = 0$$

- **Cramer's Rule:** method of using determinants to solve the linear equation

If

$$a_1x + b_1y = c_1$$

$$a_2x + b_2y = c_2$$

then

$$x = \frac{\begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}} \quad \text{and} \quad y = \frac{\begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}}$$

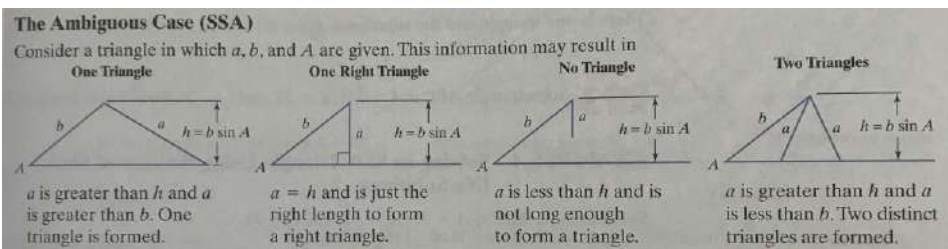
where

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \neq 0.$$

❖ Additional Topics in Trigonometry

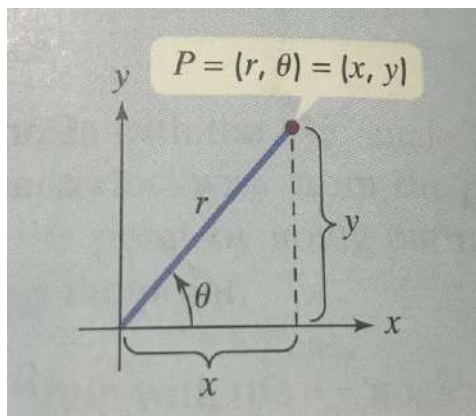
- **Law of Sines**

$$\circ \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$



- **Ambiguous Case:** given information may result in one triangle, two triangles, or no triangle at all
- **Law of Cosine**
 - $a^2 = b^2 + c^2 - 2bc \cos A$
 - $b^2 = a^2 + c^2 - 2ac \cos B$
 - $c^2 = a^2 + b^2 - 2ab \cos C$

- **Polar Coordinates:** (r, θ)
- **Rectangular Coordinates:** (x, y)
- **Multiple Representations of Points in the Coordinate System**
 - If n is any integer, the point (r, θ) can be represented as $(r, \theta) = (r, \theta + 2n\pi)$ or $(r, \theta) = (-r, \theta + \pi + 2n\pi)$
- **Relations between Polar and Rectangular Coordinates**



- $x = r \cos \theta$
- $y = r \sin \theta$
- $x^2 + y^2 = r^2$
- $\tan \theta = \frac{y}{x}$
- **Converting a Point from Rect. to Polar Coordinates** ($r > 0$ and $0 \leq \theta \leq 2\pi$)
 - 1) Plot the point (x, y)
 - 2) Find r by computing the distance from the origin to (x, y) : $r = \sqrt{x^2 + y^2}$
 - 3) Find θ using $\tan \theta = \frac{y}{x}$ with the terminal side passing through (x, y)
- **Absolute Value of a Complex Number**
 - Absolute value $a + bi$ is $|z| = |a + bi| = \sqrt{a^2 + b^2}$
- **Polar Form of a Complex Number**
 - The complex number $z = a + bi$ is written in polar form as $z = r(\cos \theta + i \sin \theta)$, where $a = r \cos \theta$, $b = r \sin \theta$, $r = \sqrt{a^2 + b^2}$, and $\tan \theta = \frac{b}{a}$.
 - **Modulus:** value of r
 - **Argument:** value of θ
- **Product of Two Complex Numbers in Polar Forms**

- Let $z_1 = \cos\theta_1 + i \sin\theta_1$ and $z_2 = \cos\theta_2 + i \sin\theta_2$, the product would be $z_1 z_2 = r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$.

■ To multiply 2 complex #s, multiply the moduli and add arguments.

- **Quotient of Two Complex Numbers in Polar Form**

- Let $z_1 = r_1(\cos\theta_1 + i \sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i \sin\theta_2)$, the quotient would be $\frac{z_1}{z_2}$.

- **DeMoivre's Theorem:** formula for the nth power and multiplying the argument by n

- Let $z = r(\cos\theta + i \sin\theta)$ be a complex number in polar form. If n is a positive integer, then z to the n th power, z^n is $z^n = [r(\cos\theta + i \sin\theta)]^n = r^n(\cos n\theta + i \sin n\theta)$.

- **Using DeMoivre's Theorem for Finding Complex Roots**

Let $w = r(\cos\theta + i \sin\theta)$ be a complex number in polar form. If $w \neq 0$, w has n distinct complex n th roots given by the formula

$$z_k = \sqrt[n]{r} \left[\cos\left(\frac{\theta + 2\pi k}{n}\right) + i \sin\left(\frac{\theta + 2\pi k}{n}\right) \right] \quad (\text{radians})$$

$$\text{or } z_k = \sqrt[n]{r} \left[\cos\left(\frac{\theta + 360^\circ k}{n}\right) + i \sin\left(\frac{\theta + 360^\circ k}{n}\right) \right] \quad (\text{degrees})$$

where $k = 0, 1, 2, \dots, n-1$.

- **Vectors:** quantities that involve both a magnitude and a direction (usually denoted with v)

- **Scalars:** quantities that involve magnitude, but no direction

- **i and j Unit Vectors**

- i - direction is along the positive x-axis
- j - direction is along the positive y-axis

- **Adding and Subtracting Vectors in Terms of i and j**

- If $a_1 i + b_1 j$ and $w = a_2 i + b_2 j$, then $v + w = (a_1 + a_2)i + (b_1 + b_2)j$ or $v - w = (a_1 - a_2)i + (b_1 - b_2)j$

- **Scalar (k) Multiplication with a Vector in Terms of i and j**

- $kv = (ka)i + (kb)j$

- **Properties of Vector Addition and Scalar Multiplication**

- **Vector Addition Properties**

■ $u + v = v + u$

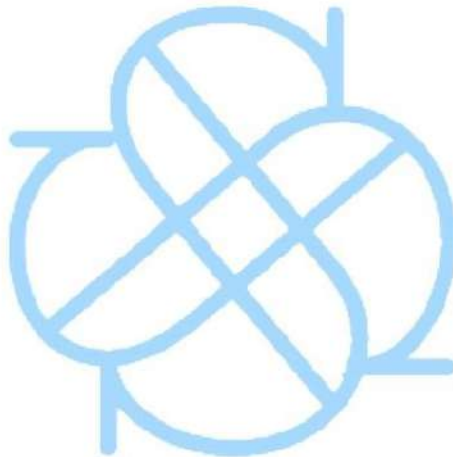
■ $(u + v) + w = u + (v + w)$

- $u + 0 = 0 + u = u$
- $u + (-u) = (-u) + u = 0$
- **Scalar Multiplication Properties**
 - $(cd)u = c(du)$
 - $c(u + v) = cu + cv$
 - $(c + d)u = cu + du$
 - $1u = u$
 - $0u = u$
 - $\|cv\| = |c|\|v\|$
- **The Dot Product:** is defined as $v \cdot w = a_1a_2 + b_1b_2$
- **Properties of the Dot Product**
 - If u, v , and w are vectors, and c is a scalar then,
 - $u \cdot v = v \cdot u$
 - $u \cdot (v + w) = u \cdot v + u \cdot w$
 - $0 \cdot v = 0$
 - $v \cdot v = \|v\|^2$
 - $(cu) \cdot v = c(u \cdot v) = u \cdot (cv)$
- **Alt. Formula for The Dot Product**
 - $v \cdot w = \|v\|\|w\|\cos \theta$
- **Vector Projection of v Onto w**
 - $proj_w v = \frac{v \cdot w}{\|w\|^2} w$

*****NOTE:** We do not claim ownership of any images used in this study guide. Some definitions, examples, and pictures were used from the following sources:

- *Blitzer Precalculus (Third Edition)*
- *Onlinemathlearning.com*
- *Basicmathematics.com*

- *Tes Tech*
- *Lumen Learning*
- *Technology UK*
- *Michael Van Biezen*
- *Mathwords*



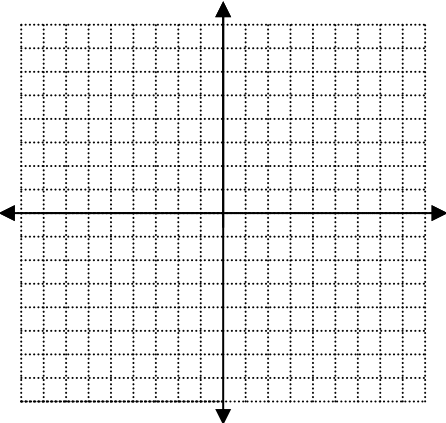
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SEMESTER 1 EXAM REVIEW

Unit 1: Intro to Pre-Calc

1. Linear Functions

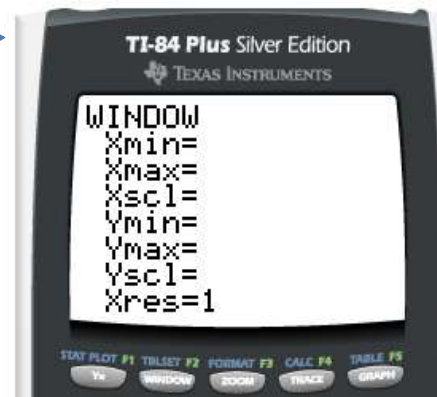
Slope Intercept Form	Standard Form	Point Slope Form
a. Write the equation of the line in slope intercept form that is perpendicular to $y = 2x - 5$ and contains $(-50, 10)$	b. Graph $3x + 2y = 10$ 	c. Write the equation of the line in point slope form that contains the points $(-50, -49)$ and $(35, 53)$

2. Regression (Best Fit Line/Curve)

The following table gives the number of motor vehicle thefts (in thousands) in the U.S. for the years 1983 - 1993. $x = 1$ represents 1983. Use the regression capabilities of your calculator to fit a cubic model to this data.

Year	1	3	4	7	8	9	10	11
Vehicle Thefts	1008	1103	1224	1565	1636	1662	1611	1561

- a. Graph the data with a friendly window. Record here $\longrightarrow$
- b. Use regression and write the equation of your model.
(Round to three decimal places)
 $V(t) =$
- c. What does $V(5.5)$ mean? Find it.
- d. Find the time(s) in which there will be 1400 thousand auto thefts.
- e. What does the y-intercept mean in this situation?
- f. Predict the auto thefts in 1995.



3. Factoring Basics: Solve the following by factoring.

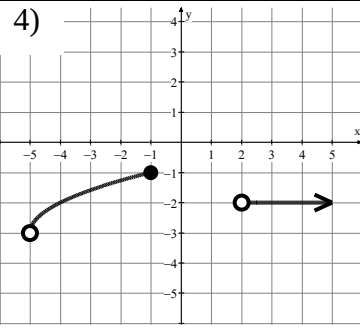
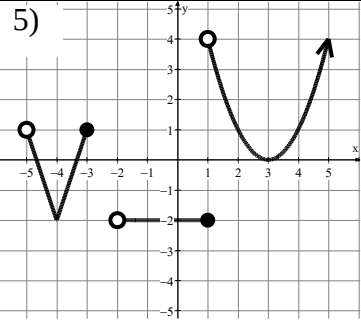
a. $x^2 - 9x = 0$

b. $x^2 - 9x - 112 = 0$

c. $2x^2 - 17x = -35$

Unit 2: Functions and Limits

For 4-5, identify the domain and range of each function. Use both interval notation and inequality notation.

4) 	Domain: Interval: _____ Inequality: _____ Range: Interval: _____ Inequality: _____	5) 	Domain: Interval: _____ Inequality: _____ Range: Interval: _____ Inequality: _____
--	--	--	--

6) Using the graph on the right, give the value of each statement.

a. $\lim_{x \rightarrow 1^-} f(x) =$

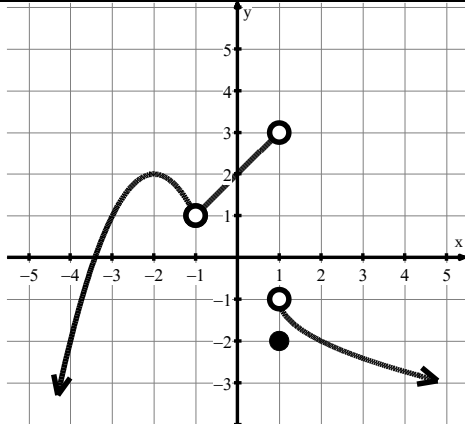
c. $\lim_{x \rightarrow -1} f(x) =$

e. $f(1) =$

b. $f(-1) =$

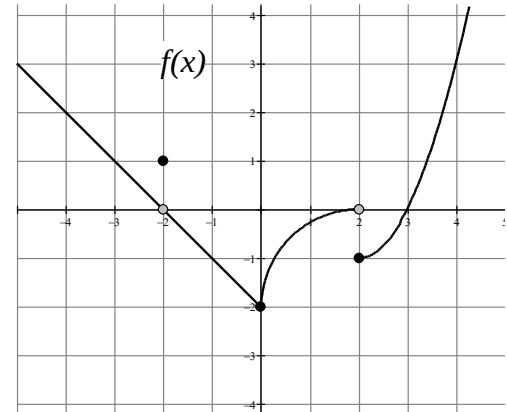
d. $\lim_{x \rightarrow -2} f(x) =$

f. $\lim_{x \rightarrow 1^+} f(x) =$



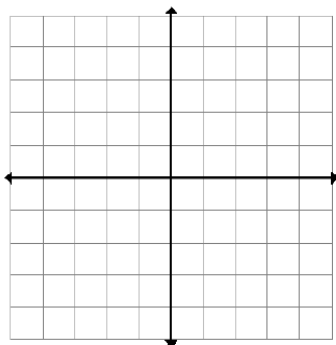
For questions 7-14, refer to the graph of $f(x)$

7. State the intervals where $f(x)$ is continuous.
8. State the values of x where the function is discontinuous and label them as removable or non-removable discontinuities.
9. $\lim_{x \rightarrow 0} f(x) =$
10. $\lim_{x \rightarrow 2^+} f(x) =$
11. $\lim_{x \rightarrow 2^-} f(x) =$
12. $\lim_{x \rightarrow 2} f(x) =$
13. $f(0) =$
14. $\lim_{x \rightarrow -2} f(x) =$

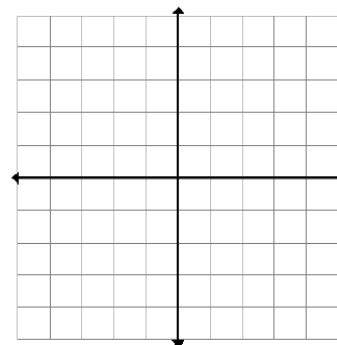


For 15-16, graph the function and determine if it has points of discontinuity. If there is a discontinuity, tell what type of discontinuity it is and its x -value. Clearly mark all asymptotes with a dotted line.

15. $f(x) = \frac{2}{x}$



16. $f(x) = \frac{x}{x^2 - 4}$



Unit 3: Function Analysis

For 17-19, find the domain of the given function. Use interval notation.

17. $f(x) = \sqrt{10-x}$

18. $f(x) = \frac{x}{x-5}$

19. $f(x) = \frac{\sqrt{x-2}}{(x+7)(x-8)}$

For 20-21, find the range of the function. Use interval notation.

20. $f(x) = (x-4)^2 + 4$

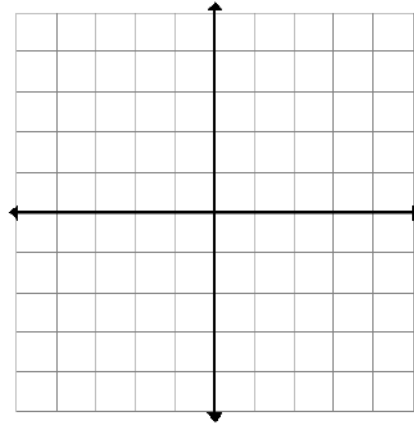
21. $f(x) = \sqrt{9+x}$

22. Sketch the piecewise function $f(x) = \begin{cases} x^3 - 1 & x \leq 0 \\ \sqrt{x} + 1 & x > 0 \end{cases}$

a. $f(2) =$

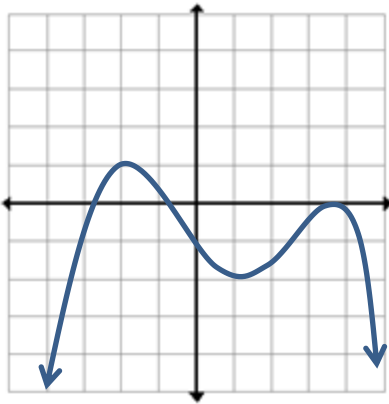
b. $f(-2) =$

c. $f(0) =$

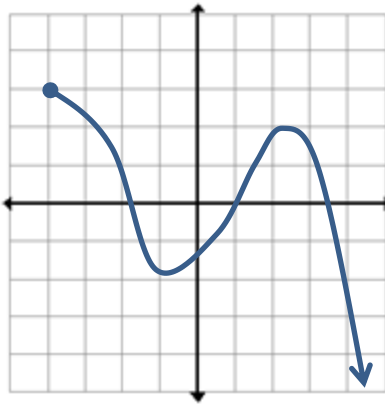


23. Label all local and absolute maximums and minimums.

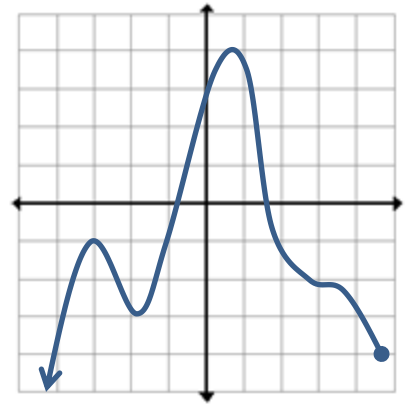
a.



b.



c.



Unit 4: Function Algebra

24. $f(x) = 3x + 11$ and $g(x) = 5x - 1$

$f \cdot g =$

25. $f(x) = 4x^2 + 2x + 3$; $g(x) = 2x - 4$

$f \circ g =$

$\frac{f}{g} =$

For 26, confirm that f and g are inverses by showing the $f(g(x)) = x$

26. $f(x) = 2x + 9$ and $g(x) = \frac{x-9}{2}$

For 27-32, if $f(x) = 2x - 5$ and the $g(x) = x^2 + 2x - 3$, find the following...

27. $f(2)$

28. $g(-2)$

29. $f(g(0))$

30. $f - g$

31. $f(x + h)$

32. $(f + g)(2)$

33. Is $f(x) = \frac{x+1}{x^2-1}$ even, odd, or neither. Justify your answer!

Transformations

34. $y = 2(x - 5)^3 - 4$

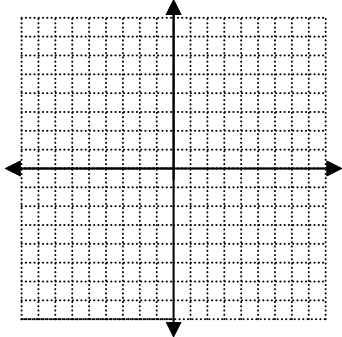
Name function: _____

Translation:

Scale:

Reflection:

SKETCH GRAPH!



35. $f(x) = -|3x + 6| + 5$

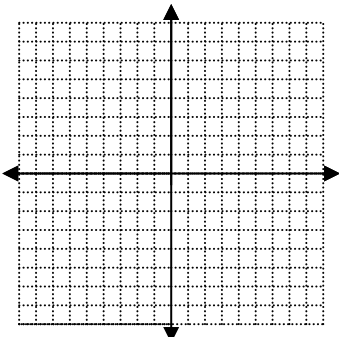
Name function: _____

Translation:

Scale:

Reflection:

SKETCH GRAPH!



36. $y = \sqrt{-x} + 3$

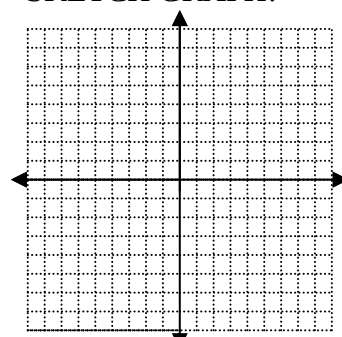
Name function: _____

Translation:

Scale:

Reflection:

SKETCH GRAPH!



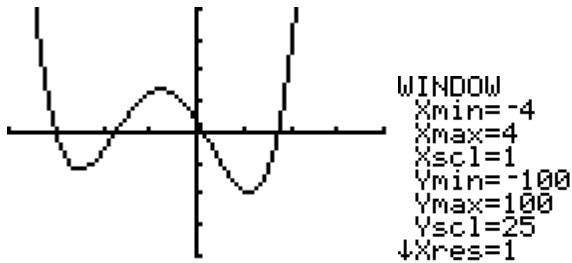
Unit 5: Polynomials

Graph on your calculator to solve the following.

37. $0 = x^4 + 2x^2 - 3x - 1$

38. $3x^6 - 2x^5 = 8 - 3x^2$

39. Use the graph of the function to determine at least one zero, then find the exact values of all the zeros using the Factor Theorem. $f(x) = 7x^4 + 20x^3 - 24x^2 - 60x + 9$



40. Factor the following.

a. $6x^2 + 13xy - 5y^2$

b. $5x^3 - 30x^2 - 8x + 48$

c. $216 + x^3$

41. Factor to solve the following.

a. $4x^4 + 64x = 0$

b. $x^3 - 6x^2 + 8x = 0$

c. $x^4 - 11x^2 = -30$

Unit 6: Rational Functions

42. $f(x) = \frac{4}{x^3 - 9x}$

Vertical Asymptotes/Holes:

x-intercepts:

y-intercepts:

Horizontal/Slant Asymptotes:

Solve:

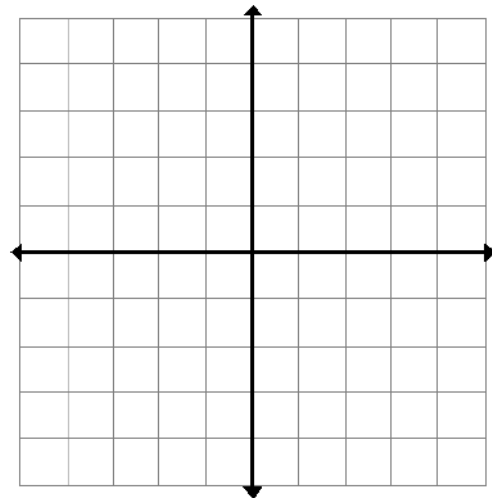
43. $\frac{2x}{x+2} = \frac{5}{x^2-x-6} - \frac{1}{x-3}$

44. $\frac{7n^2-n}{n+9} = 2n - 1$

Simplify:

45. $\frac{\frac{3g}{g-5} + 2}{\frac{3}{g-5} - 4}$

46. $\frac{x+4}{\sqrt{x} - \sqrt{x+2}}$



Unit 7: Exponential and Logarithmic Functions

Evaluate.

47. $\log_2 64$

48. $\log_6 \frac{1}{36}$

Solve for the indicated variable. Round to three digits where applicable.

49. $(5^{2x})^{(x+7)} = 1$

50. $56 \log_7 x = 203$

51. $e^{2x} - 22 = 25$

52. $\ln x = \ln (x + 14) - \ln (x + 90)$

53. $5 \log_9 (x - 12) = 45$

Find the inverse of the given function.

54. $f(x) = 8^{x-3}$

55. $\ln y = 12 \ln (x - 3)$

56. At what rate compounded continuously will \$12,000 have to be invested to amount to \$35,000 in 12 years?
Use $A = Pe^{rt}$.

APPLICATIONS

The formula for the path of a flying bullet is given: $h = -9.8t^2 + vt + s$ where h = height of object, in meters
 t = time, in seconds, v = velocity, in meters per second and s = starting height, in meters

Bob shoots a gun straight up with a velocity of 300 meters per second and a starting height of 2 meters.

57. What is the equation that represents this situation? $y =$ _____

58. What do the axes represent? x -axis: _____

y -axis: _____

59. What does the y -intercept represent to Bob? _____

60. What do the x -intercepts represent to Bob? _____

61. How high is the bullet after 4 seconds? _____

62. How long will it take for the bullet to hit the ground after it is fired? _____

63. What is the maximum height of the bullet? _____

64. At what time(s) will the bullet be 1200 meters in the air? _____

65. The concentration C (in milligrams) of a certain drug in a patient's bloodstream t minutes after injection is given by $C(t) = \frac{50t}{(t^2 + 25)}$. Using your graphing calculator, graph $C(t)$ to answer the following questions.

- What happens to the concentration of the drug as t increases?
- Determine the time at which the concentration is highest.
- What is the highest concentration?
- What is the horizontal asymptote of $C(t)$? What does it mean in this situation?
- If the drug is to be re-administered when the concentration decreases to 2 milligrams, how many minutes after the initial dose is the drug re-administered?